3

The Semiparametric and Nonparametric Cure Models

3.1 Introduction

Parametric cure models introduced in the last chapter are attractive in their relatively simple estimation methods, and they are usually very efficient and allow for extrapolation if the parametric assumptions are acceptable. However, verifying the parametric assumptions, particularly the assumptions for the latency distribution, can be challenging. The strong parametric assumptions increase the risk of misspecification of the cure model for a given data, which can lead to biased estimates. Semiparametric and nonparametric cure models are proposed to relax the assumptions so that the results are less sensitive to the model assumptions. We first introduce semiparametric mixture cure models in Section 3.2, which do not make parametric assumptions in the baseline distribution of the latency submodel. Fully nonparametric approaches are introduced in Section 3.3. Section 3.4 presents semiparametric estimation methods for some non-mixture cure models. Section 3.5 discusses some model assessment methods for semiparametric and nonparametric cure methods, followed by Section 3.6 on available software for the some discussed models and their applications to real data. This chapter also primarily focuses on independent and right-censored data.

3.2 Semiparametric Mixture Cure Models

Both the incidence and latency parts of a mixture cure model can be modeled with weaker and semiparametric or nonparametric assumptions. In this section, we focus on the latency part of the mixture model only and introduce a few semiparametric approaches. The incidence part of the mixture cure model is specified in the same way as described in Section 2.2.1.

3.2.1 Semiparametric PH Latency Submodel

If the effects of $\boldsymbol{x}$ on $S_u(t)$ is modeled using the PH assumption in (2.5) and the baseline survival function $S_u(\cdot)$ is not parametrically specified, we have a semiparametric PH model for the latency part. Due to the unspecified baseline survival function, the direct maximization of the likelihood function (2.12) in Section 2.3.1 becomes challenging (Lu, 2008). We demonstrate how the EM algorithm in Section 2.3.2 becomes a better way to estimate the model, as proposed by many researchers (Peng and Dear, 2000; Sy and Taylor, 2000; Peng, 2003b).

Under the PH latency submodel (2.5), the corresponding hazard function is $h_u(t|\boldsymbol{x}) = h_{u0}(t)e^{\boldsymbol{\beta}'\boldsymbol{x}}$, and (2.16) becomes

$$\ell_2(\boldsymbol{\beta}, S_{u0}) = \sum_{i=1}^{n} w_i \left\{\delta_i \log h_{u0}(t_i) + \delta_i \boldsymbol{\beta}'\boldsymbol{x}_i + e^{\boldsymbol{\beta}'\boldsymbol{x}_i} \log[S_{u0}(t_i)]\right\} \tag{3.1}$$

$$= \sum_{i=1}^{n} \left\{\delta_i \log h_{u0}(t_i) + \delta_i \boldsymbol{\beta}'\boldsymbol{x}_i + e^{\log w_i + \boldsymbol{\beta}'\boldsymbol{x}_i} \log[S_{u0}(t_i)]\right\} \tag{3.2}$$

where the fact that $w_i = 1$ and $\log w_i = 0$ for $\delta_i = 1$ is used to obtain (3.2). As discussed in Section 2.3.2, (3.1) can be viewed as a weighted likelihood function from a PH model for all the subjects with $w_i > 0$. On the other hand, (3.2) can be viewed as the log-likelihood function from the PH model with $\log w_i$ as an offset term. Therefore, $\boldsymbol{\beta}$ and $S_{u0}(\cdot)$ can be updated either by the Newton-Raphson method or by an existing software package for the classical PH model that accepts either case weights or offset terms.

If $S_{u0}(t)$ is unspecified or nonparametrically specified, it is easy to see that (3.1) or (3.2) becomes the log-likelihood function considered in the context of the Cox PH model with the weights w_i or the offset terms $\log w_i$, which suggests that $\boldsymbol{\beta}$ and S_{u0} be updated separately as in Cox's PH model (Peng and Dear, 2000), where $\boldsymbol{\beta}$ is updated via the following partial log-likelihood function corresponding to (3.2)

$$\log \prod_{j=1}^{k} \frac{\exp(\boldsymbol{\beta}'\boldsymbol{s}_j)}{\left\{\sum_{i\in R_j} \exp(\log w_i + \boldsymbol{\beta}'\boldsymbol{x}_i)\right\}^{d_j}},$$

k is the number of distinct uncensored times $\tau_1 < \tau_2 < \cdots \tau_k$ in the data, $\boldsymbol{s}_j = \sum_{i:t_i=\tau_j} \delta_i \boldsymbol{x}_i$, d_j is the number of uncensored times equal to τ_j, and R_j is the risk set at τ_j. With updated $\hat{\boldsymbol{\beta}}$, the baseline survival function $S_{u0}(t)$ can be estimated with the Nelson-Aalen estimator

$$\hat{S}_{u0}(t) = \exp\left(-\sum_{j:\tau_j<t} \frac{d_j}{\sum_{i\in R_j} \exp(\log w_i + \hat{\boldsymbol{\beta}}'\boldsymbol{x}_i)}\right). \tag{3.3}$$

This approach can be easily implemented using existing software for Cox's

PH model. The M-step also maximizes (2.15), which can be carried out as described in Section 2.3.2.

Another approach is to use a discrete distribution that has probability mass on the distinct uncensored times only to approximate the failure time distribution of $T|Y = 1$ (Sy and Taylor, 2000). The likelihood function (3.2) reduces to

$$\log \prod_{j=1}^{k} \left[\prod_{i \in D_j} h_{u0}(\tau_j) e^{\boldsymbol{\beta}' \boldsymbol{x}_i} S_{u0}(\tau_j -)^{\exp(\boldsymbol{\beta}' \boldsymbol{x}_i)} \prod_{i \in C_j} S_{u0}(\tau_j)^{w_i \exp(\boldsymbol{\beta}' \boldsymbol{x}_i)} \right]$$

which can be readily maximized using the Newton-Raphson method. However few existing programs for Cox's PH model are readily available to maximize this log likelihood and a specialized software program is required.

The standard errors of the estimates can be obtained using the method of Louis (1982) after the EM algorithm converges if the approach of Sy and Taylor (2000) is used in the M-step. See Sy and Taylor (2001) for details. If the approach of Peng and Dear (2000) is used in the M-step, a bootstrap method may be used to estimate the standard errors of the estimates.

The methods above only produce a nonparametric non-smooth estimate of the baseline survival function $S_{u0}(t)$. If a smooth baseline survival function is preferred, the PH mixture cure model can be estimated by allowing flexible modeling of the hazard function $h_{u0}(t)$ using M-splines (Corbiere et al., 2009). The smoothing parameter in the M-splines can be determined using cross-validation.

3.2.1.1 Restrictions on the Upper Tail of the Baseline Distribution

Due to the presence of substantial censoring after τ_k, the nonparametric estimate of the baseline survival function $\hat{S}_{u0}(t) \not\to 0$ when $t \geq \tau_k$. Even though this is not an issue in the standard PH model, it causes an identifiability issue in the cure model context: It essentially implies that there is a non-zero probability for an uncured subject to never have any event. To alleviate this identifiability issue, $\hat{S}_{u0}(t)$ should go to 0 soon after τ_k if there is sufficient follow-up in a study. A popular approach is to impose a zero tail restriction on the survival estimate (Taylor, 1995). That is, $\hat{S}_{u0}(t) = 0$ for any $t \geq \tau_k$. This approach is simple to use, supported by other studies such as Maller and Zhou (1992, 1994), and is widely used in other semiparametric cure models in the following sections. Other more complicated approaches can be considered to allow $\hat{S}_{u0}(t)$ to go to zero gradually from $\hat{S}_{u0}(\tau_k)$, for example, at the same rate as an exponential survival function (ETAIL) or as a Weibull survival function (WTAIL) (Peng, 2003a), an idea similar to that used by Moeschberger and Klein (1985) for the Kaplan-Meier survival estimator.

3.2.1.2 Time-Dependent Covariates in the Latency Submodel

Time-dependent covariates can be conveniently included in Cox's PH model. They can also be considered in the latency submodel of the semiparametric PH mixture cure model (Dirick et al., 2017). Assume that the covariates in $\boldsymbol{x}$ are allowed to be time-dependent and denoted as $\boldsymbol{x}(t)$. Then the latency submodel can be written as

$$h_u(t|\boldsymbol{x}) = h_{u0}(t)e^{\boldsymbol{x}(t)'\boldsymbol{\beta}}$$

The corresponding cumulative hazard function is:

$$H_u(t|\boldsymbol{x}\{t\}) = \int_0^t h_{u0}(s)e^{\boldsymbol{\beta}'\boldsymbol{x}(s)}ds \tag{3.4}$$

where $\boldsymbol{x}\{t\} = \{\boldsymbol{x}(s); 0 < s \leq t\}$ is the history of $\boldsymbol{x}$ up to t. It is worth to point out that $S_u(t|\boldsymbol{x}\{t\}) = \exp[-H_u(t|\boldsymbol{x}\{t\})]$ and the effects of $\boldsymbol{x}$ cannot be factored out from the integration as in the usual PH model due to the presence of the time-dependent covariates, and estimating the baseline hazard function $h_{u0}(t)$ or the cumulative baseline hazard function $H_{u0}(t)$ is no longer useful. Given the current estimates $\hat{\boldsymbol{\beta}}$, a Nelson-Aalen estimator of the cumulative hazard function $H_u(t|\boldsymbol{x}\{t\})$ can be considered as follows (Dong et al., 2020):

$$\hat{H}_u(t|\boldsymbol{x}\{t\}) = \sum_{i=1}^n \int_0^t \frac{e^{\hat{\boldsymbol{\beta}}'\boldsymbol{x}(s)}dN_i(s)}{\sum_{j=1}^n Y_j(s)w_j e^{\hat{\boldsymbol{\beta}}'\boldsymbol{x}_j(s)}} \tag{3.5}$$

and the corresponding survival estimator is $\hat{S}_u(t|\boldsymbol{x}\{t\}) = \exp[-\hat{H}_u(t|\boldsymbol{x}\{t\})]$. As suggested by Thomas and Reyes (2014) for the PH model, the cumulative hazard function above can be readily obtained using `PHREG` in SAS if the following form is considered:

$$\hat{H}_u(t|\boldsymbol{x}\{t\}) = \sum_{i=1}^n \int_0^t \frac{dN_i(s)}{\sum_{j=1}^n Y_j(s)w_j e^{\hat{\boldsymbol{\beta}}'[\boldsymbol{x}_j(s)-\boldsymbol{x}(s)]}} \tag{3.6}$$

which implies that the cumulative hazard function can be obtained if $\boldsymbol{x}_j(t)$ is replaced with $\boldsymbol{x}_j(t) - \boldsymbol{x}(t)$.

Generally, there are two classes of time-dependent covariates, internal time-dependent covariates and external time-dependent covariates (Kalbfleisch and Prentice, 2002). The observation of an internal time-dependent covariates requires that the study subject be at risk while this is not required for an external time-dependent covariate. Thus, caution should be exercised when using internal time-dependent covariates in the latency submodel.

3.2.2 Semiparametric AFT Latency Submodel

If the effects of $\boldsymbol{x}$ on $S_u(t)$ is specified under the AFT in (2.6) and the baseline survival function $S_{u0}(t)$ is not parametrically specified, we have a semiparametric AFT model for the latency part. Similarly, due to the unspecified baseline survival function, the direct maximization of the likelihood function (2.12)

in Section 2.3.1 becomes infeasible and the EM algorithm in Section 2.3.2 is often used to estimate the model, as suggested by many researchers (Li and Taylor, 2002a,b; Zhang and Peng, 2007; Lu, 2010). Even though (2.16) in theory can be updated with a program for the semiparametric AFT model that takes case weights, such a program, unlike those for the semiparametric PH model, is still rare and not available in mainstream statistical software packages. We will introduce a few proposed semiparametric methods to update $\boldsymbol{\beta}$ and $S_{u0}(t)$ in the AFT latency submodel.

3.2.2.1 Linear Rank Method

Let $f_0(\cdot)$, $h_0(\cdot)$, and $S_0(\cdot)$ be the density, hazard, and survival functions of ϵ in (2.7). The log-likelihood function (2.16) can be written as a function of $f_0(\cdot)$, $h_0(\cdot)$, and $S_0(\cdot)$ as follows:

$$\begin{aligned}\ell_2(\boldsymbol{\beta},\boldsymbol{\alpha}) &= \sum_{i=1,w_i>0}^{n} \left\{\delta_i \log f_0(\log t_i - \boldsymbol{\beta}'\boldsymbol{x}_i) + w_i(1-\delta_i)\log[S_0(\log t_i - \boldsymbol{\beta}'\boldsymbol{x}_i)]\right\} \\ &= \sum_{i=1,w_i>0}^{n} \left\{\delta_i \log h_0(\log t_i - \boldsymbol{\beta}'\boldsymbol{x}_i) + w_i\log[S_0(\log t_i - \boldsymbol{\beta}'\boldsymbol{x}_i)]\right\} \quad (3.7)\end{aligned}$$

The following rank-like estimating function can be used to update $\boldsymbol{\beta}$ in the above likelihood function (Zhang and Peng, 2007):

$$\sum_{i=1}^{n} \delta_i g(\epsilon_i)\left(\boldsymbol{x}_i - \frac{\sum_{j=1}^{n}\boldsymbol{x}_j w_j I(\epsilon_j \geq \epsilon_i)}{\sum_{j=1}^{n} w_j I(\epsilon_j \geq \epsilon_i)}\right),$$

where $\epsilon_i = \log t_i - \boldsymbol{\beta}'\boldsymbol{x}_i$ and $g(\cdot)$ is a weight function. A Gehan-type weight function $g(u) = \sum_{j=1}^{n} I(\epsilon_j \geq u)w_j/n$ will further simplify the above estimating function into

$$n^{-1}\sum_{i=1}^{n}\sum_{j=1}^{n}\delta_i(\boldsymbol{x}_i - \boldsymbol{x}_j)w_j I(\epsilon_j \geq \epsilon_i),$$

which can be viewed as a gradient of a convex function that can be minimized by the linear programming method (Jin et al., 2003).

Given estimated $\hat{\boldsymbol{\beta}}$, the survival function of ϵ can also be estimated nonparametrically using the Nelson-Aalen estimator:

$$\hat{S}_0(\epsilon) = \exp\left(-\sum_{j:\tau_j^*<\epsilon}\frac{d_j}{\sum_{i\in R_j} w_i}\right), \quad (3.8)$$

where $\tau_1^* < \tau_2^* < \cdots < \tau_k^*$ are distinct uncensored values of $\log t_i - \boldsymbol{\beta}'\boldsymbol{x}_i$ and d_j and R_j are the corresponding number of uncensored times and the risk set at τ_j^*. A zero-tail restriction discussed in Section 3.2.1.1 is recommended for this estimator to alleviate the identifiability issue. An estimate of $S_{u0}(t)$ can be obtained easily from $\hat{S}_0(\epsilon)$.

3.2.2.2 M-Estimation Method

Another semiparametric approach for the semiparametric model in the AFT latency part is based on the score function of the likelihood function (3.7) (Li and Taylor, 2002b):

$$\sum_{i=1,w_i>0}^{n} \boldsymbol{x}_i \left\{ -\delta_i \frac{f_0'}{f_0}(\log t_i - \boldsymbol{\beta}'\boldsymbol{x}_i) + w_i(1-\delta_i)\frac{f_0}{S_0}(\log t_i - \boldsymbol{\beta}'\boldsymbol{x}_i) \right\},$$

which can be written into the following general estimating equation for $\boldsymbol{\beta}$:

$$\sum_{i=1,w_i>0}^{n} \boldsymbol{x}_i \left\{ -\delta_i s(\log t_i - \boldsymbol{\beta}'\boldsymbol{x}_i) - w_i(1-\delta_i)\frac{\int_{\log t_i - \boldsymbol{\beta}'\boldsymbol{x}_i}^{\infty} s(u)dS_0(u)}{S_0(\log t_i - \boldsymbol{\beta}'\boldsymbol{x}_i)} \right\},$$

where $s(u)$ is a reasonable score function, such as $s(u) = u$. For the unknown survival function $S_0(t)$, it can be estimated by (3.8).

This method and the linear rank method in the last section can be viewed as extensions of the semiparametric methods proposed for the AFT model (Ritov, 1990), and the two methods are asymptotically equivalent (Zhang and Peng, 2012). However, both methods involve non-smooth estimating equations, which can be challenging in implementing the methods. The following method will address this issue directly.

3.2.2.3 Kernel Smoothing Method

To avoid the non-smoothed likelihood functions and estimating equations, a semiparametric kernel smoothed likelihood function can be used to approximate the likelihood function (2.16) (Lu, 2010; Zeng and Lin, 2007), which leads to the following likelihood function for $\boldsymbol{\beta}$:

$$\frac{1}{n}\sum_{i=1}^{n}\delta_i \log\left[\frac{1}{n}\sum_{j=1}^{n}\delta_j K_h(\log t_j - \log t_i + \boldsymbol{\beta}'\boldsymbol{x}_i - \boldsymbol{\beta}'\boldsymbol{x}_j)\right]$$
$$-\frac{1}{n}\sum_{i=1}^{n}\delta_i \log\left[\frac{1}{n}\sum_{j=1}^{n} w_j \int_{-\infty}^{\log t_j - \log t_i + \boldsymbol{\beta}'\boldsymbol{x}_i - \boldsymbol{\beta}'\boldsymbol{x}_j} K_h(u)du\right],$$

where $K_h(u) = K(u/h)/h$ is a kernel function with bandwidth h and a symmetric function $K(u)$. The examples of the kernel function include (Härdle et al., 2004):

- Gaussian density kernel: $K(x) = \frac{1}{\sqrt{2\pi}}\exp\left(-\frac{1}{2}x^2\right)$,
- cosine kernel: $K(x) = \frac{\pi}{4}\cos\left(\frac{\pi}{2}x\right)$ for $|x| \le 1$,
- tri-weight kernel: $K(x) = \frac{35}{32}(1-x^2)^3$ for $|x| \le 1$,

- triangular kernel: $K(x) = (1 - |x|)$ for $|x| \leq 1$.

The resulting likelihood function is a smooth function of $\boldsymbol{\beta}$ and can be maximized readily using the Newton-Raphson method. With the estimated $\hat{\boldsymbol{\beta}}$, the baseline hazard function can be estimated by a smoothed estimator

$$\hat{h}_{u0}(t) = \frac{t^{-1} \sum_{i=1}^{n} \delta_i K_h(\log t_i - \hat{\boldsymbol{\beta}}' \boldsymbol{x}_i - \log t)}{\sum_{i=1}^{n} w_i \int_{-\infty}^{\log t_i - \hat{\boldsymbol{\beta}}' \boldsymbol{x}_i - \log t} K_h(u) du}.$$

The estimates from the smoothed likelihood function can achieve semiparametric efficiency bound for $\boldsymbol{\beta}$ and $\boldsymbol{\gamma}$ in theory. Practically, the results of the kernel method depend on the value of the bandwidth h. Lu (2010) suggested two optimal bandwidth formulas: $(8\sqrt{2}/3)^{1/5}\sigma_1 n^{-1/5}$ and $4^{1/3}\sigma_2 n^{-1/3}$, where σ_1 is the sample standard deviation of the $(\log t_i - \boldsymbol{\beta}'_{1,ls}\boldsymbol{x}_i)$ for uncensored data with $\boldsymbol{\beta}_{1,ls}$ the least square estimate of $\boldsymbol{\beta}$ using only uncensored data, while σ_2 is the sample standard deviation of the $(\log t_i - \boldsymbol{\beta}'_{2,ls}\boldsymbol{x}_i)$ for all the data with $\boldsymbol{\beta}_{2,ls}$ the corresponding least square estimate of $\boldsymbol{\beta}$.

It is worth to note that the zero-tail restriction discussed in Section 3.2.1.1 should be considered for all the methods proposed for the semiparametric AFT mixture cure model to reduce the identifiability issue.

3.2.3 Semiparametric AH Latency Submodel

When the latency survival function $S_u(t|\boldsymbol{x})$ follows the AH model (2.10) and the baseline survival function is not parametrically specified, the direct maximization method is not easy to use as in the previous semiparametric models, and the EM algorithm is preferred. We introduce two semiparametric methods based on the EM algorithm to estimate the parameters in the latency submodel.

3.2.3.1 Linear Rank Method

One method is to use a rank-type estimating equation to update $\boldsymbol{\beta}$ (Zhang and Peng, 2009a). This method is based on the fact that under the AH model, (2.16) can be written as

$$\ell_2(\boldsymbol{\beta}, \boldsymbol{\alpha}) = \sum_{i=1, w_i>0}^{n} \left\{ \delta_i \log \left[w_i h_{u0}(t_i e^{\boldsymbol{\beta}' \boldsymbol{x}_i}) \right] + \log \left[S_{u0}(t_i e^{\boldsymbol{\beta}' \boldsymbol{x}_i})^{w_i \exp(-\boldsymbol{\beta}' \boldsymbol{x}_i)} \right] \right\}, \tag{3.9}$$

which can be viewed as the log-likelihood function of the AH model with the hazard function $w_i h_{u0}(t_i e^{\boldsymbol{\beta}' \boldsymbol{x}_i})$. Thus following Chen and Wang (2000) and Chen (2001), a rank-type estimating equation for $\boldsymbol{\beta}$ can be constructed as follows:

$$\sum_{i=1}^{n} \delta_i g(t_i e^{\boldsymbol{\beta}' \boldsymbol{x}_i}) \left(\boldsymbol{x}_i - \frac{\sum_{j=1}^{n} \boldsymbol{x}_j w_j e^{-\boldsymbol{\beta}' \boldsymbol{x}_j} I(t_j e^{\boldsymbol{\beta}' \boldsymbol{x}_j} \geq t_i e^{\boldsymbol{\beta}' \boldsymbol{x}_i})}{\sum_{j=1}^{n} w_j e^{-\boldsymbol{\beta}' \boldsymbol{x}_j} I(t_j e^{\boldsymbol{\beta}' \boldsymbol{x}_j} \geq t_i e^{\boldsymbol{\beta}' \boldsymbol{x}_i})} \right), \tag{3.10}$$

with $g(\cdot)$ as a weight function. It can be further simplified as

$$\frac{1}{n}\sum_{i=1}^{n}\sum_{j=1}^{n}\delta_i(\boldsymbol{x}_i-\boldsymbol{x}_j)w_j e^{-\boldsymbol{\beta}'\boldsymbol{x}_j}I(t_j e^{\boldsymbol{\beta}'\boldsymbol{x}_j}\geq t_i e^{\boldsymbol{\beta}'\boldsymbol{x}_i}),$$

if $g(u)=\frac{1}{n}\sum_{j=1}^{n}w_j e^{-\boldsymbol{\beta}'\boldsymbol{x}_j}I(t_j e^{\boldsymbol{\beta}'\boldsymbol{x}_j}\geq u)$. With an updated $\hat{\boldsymbol{\beta}}$ from the estimating equation, the baseline cumulative hazard function estimate can be obtained as

$$\hat{H}_{u0}(t)=\sum_{j:\tau_j<t}\frac{d_j}{\sum_{i\in R_j}w_i\exp(-\hat{\boldsymbol{\beta}}'\boldsymbol{x}_i)}.$$

This will complete the M-step and the EM algorithm in Section 2.3.2 can proceed.

3.2.3.2 Kernel Smoothing Method

The linear rank method suffers the same computational issues as the one for the AFT mixture cure model due to the non-smooth estimating equation (3.10). One can use a kernel smoothing method similar to the one considered for the AFT mixture cure model in Section 3.2.2.3 to estimate the parameters in the semiparametric AH mixture cure model (Zhang et al., 2013). Under this method, a kernel smoothed likelihood function that approximates (3.9) is

$$-\frac{1}{n}\sum_{i=1}^{n}\delta_i\boldsymbol{\beta}'\boldsymbol{x}_i+\frac{1}{n}\sum_{i=1}^{n}\delta_i\log\left[\frac{1}{n}\sum_{j=1}^{n}\delta_j K_h(\log t_j-\log t_i+\boldsymbol{\beta}'\boldsymbol{x}_j-\boldsymbol{\beta}'\boldsymbol{x}_i)\right]$$
$$-\frac{1}{n}\sum_{i=1}^{n}\delta_i\log\left[\frac{1}{n}\sum_{j=1}^{n}w_j\int_{-\infty}^{\log t_j-\log t_i+\boldsymbol{\beta}'\boldsymbol{x}_j-\boldsymbol{\beta}'\boldsymbol{x}_i}e^{-\boldsymbol{\beta}'\boldsymbol{x}_j}K_h(u)du\right].$$

This function can be maximized via the Newton-Raphson method to obtain an updated estimate $\hat{\boldsymbol{\beta}}$, with which the baseline hazard function can be estimated by

$$\hat{h}_{u0}(t)=\frac{t^{-1}\sum_{i=1}^{n}\delta_i K_h(\log t_i+\hat{\boldsymbol{\beta}}'\boldsymbol{x}_i-\log t)}{\sum_{i=1}^{n}w_i e^{-\hat{\boldsymbol{\beta}}'\boldsymbol{x}_i}\int_{-\infty}^{\log t_i+\hat{\boldsymbol{\beta}}'\boldsymbol{x}_i-\log t}K_h(u)du}.$$

The bandwidth h can be chosen from $\sigma n^{-1/3}$, $\sigma n^{-1/5}$ and $\sigma n^{-1/7}$, where σ is the sample standard deviation of $\log t_i$.

Similar to the semiparametric AFT mixture cure model, when obtaining the corresponding baseline survival function $S_{u0}(t)$ in all the methods above, the zero-tail restriction as in Section 3.2.1.1 should be considered to alleviate the identifiability issue.

3.2.4 Semiparametric Transformation Latency Submodels

The latency submodel can also be specified using the linear transformation model as follows (Lu and Ying, 2004):

$$g(T) = -\boldsymbol{\beta}'\boldsymbol{x} + \epsilon \tag{3.11}$$

where $g(\cdot)$ is a monotone increasing function with $g(0) = -\infty$ and ϵ is the error term with a known continuous distribution that is independent of $\boldsymbol{x}$ and survival time. Let the survival function of ϵ be $S_\epsilon(t)$. This latency submodel is equivalent to

$$S_u(t|\boldsymbol{x}) = S_\epsilon[g(t) + \boldsymbol{\beta}'\boldsymbol{x}]. \tag{3.12}$$

If ϵ is chosen to follow the extreme value distribution with the survival function $S_\epsilon(t) = \exp(-e^t)$, then $S_u(t|\boldsymbol{x}) = \exp(-e^{g(t)}e^{\boldsymbol{\beta}'\boldsymbol{x}})$. This is a PH model (2.5) with $e^{g(t)}$ as the baseline cumulative hazard function for uncured subjects. On the other hand, if ϵ follows the logistic distribution with the survival function $S_\epsilon(t) = 1/(1+e^t)$, then $S_u(t|\boldsymbol{x}) = 1/(1+e^{g(t)}e^{\boldsymbol{x}'\beta})$. This is a PO model (2.11) with $e^{g(t)}$ as the baseline odds function.

This model can be estimated semiparametrically without making a parametric assumption for $g(\cdot)$. The following estimating equations can be used to estimate $\boldsymbol{\gamma}$, $\boldsymbol{\beta}$ and $g(t)$ respectively (Lu and Ying, 2004)

$$\sum_{i=1}^{n} \boldsymbol{z}_i \left\{\delta_i + (1-\delta_i)w_{0i}(t_i) - \pi(\boldsymbol{z}_i)\right\} = 0 \tag{3.13}$$

$$\sum_{i=1}^{n} \int_0^\infty \boldsymbol{x}_i[dN_i(t) - Y_i(t)d\log(1-w_{0i}(t))] = 0 \tag{3.14}$$

$$\sum_{i=1}^{n} [dN_i(t) - Y_i(t)d\log(1-w_{0i}(t))] = 0 \tag{3.15}$$

where $N_i(t) = \delta_i I(t_i \le t)$, $Y_i(t) = I(t_i \ge t)$, and $w_{0i}(t)$ is defined in (2.13). It is easy to see that (3.13) is equivalent to the estimating equation for (2.15), and (3.14) and (3.15) are obtained from the fact that a process defined as

$$M(t) = N(t) + \int_0^t Y(s)d\log\left[1 + e^{\boldsymbol{\gamma}'\boldsymbol{z} - H_u(s|\boldsymbol{x})}\right] \tag{3.16}$$

is a martingale process. The transformation function $g(\cdot)$ can be estimated within a class of nondecreasing step functions on $[0,\infty)$ with $g(0) = -\infty$ and with jumps only at the observed failure times $\tau_1, \ldots, \tau_k$. In this case, the estimating equations for $\boldsymbol{\beta}$ and $g(\tau_j)$, $j = 1, \ldots, k$, can be further simplified

into

$$\sum_{i=1}^{n} \boldsymbol{x}_i[\delta_i - \log(1 - w_{0i}(\min(t_i, \tau_k)))] = 0 \tag{3.17}$$

$$d_j + \sum_{i=1}^{n} Y_i(\tau_j) \log \frac{1 - w_{0i}(\tau_j -)}{1 - w_{0i}(\tau_j)} = 0, \quad j = 1, \ldots, k \tag{3.18}$$

For any specific distribution of ϵ, $\boldsymbol{\gamma}$, $\boldsymbol{\beta}$ and $g(\tau_j)$, $j = 1, \ldots, k$, can be estimated by solving (3.13), (3.17), and (3.18) iteratively.

This model was further considered for a case with time-dependent covariates and dependent censoring (Othus et al., 2009). They proposed to use an inverse censoring probability reweighting scheme to deal with the dependent censoring and to construct unbiased estimating equations to estimate the parameters.

Furthermore, it is possible to specify the distribution of ϵ in (3.11) with an additional parameter so that the resulting latency submodel can unify the PH and PO latency submodels that are produced by $S_\epsilon(t) = \exp(-e^t)$ and $S_\epsilon(t) = 1/(1 + e^t)$ respectively. For example, if

$$S_\epsilon(t) = \exp([1 - (1 + e^t)^\rho]/\rho) \tag{3.19}$$

for $\rho \geq 0$ or if

$$S_\epsilon(t) = (1 + \rho e^t)^{-1/\rho} \tag{3.20}$$

for $\rho \geq 0$, then $S_\epsilon(t) = 1/(1 + e^t)$ when $\rho = 0$ in (3.19) (or $\rho = 1$ in (3.20)) and $S_\epsilon(t) = \exp(-e^t)$ when $\rho = 1$ in (3.19) (or $\rho = 0$ in (3.20)). Thus, the corresponding latency submodel unifies the PH and the PO latency submodels. Suppose that $S_\epsilon(t)$ is given in (3.20). Following (3.12), the latency survival function is

$$S_u(t|\boldsymbol{x}) = \left(1 + \rho e^{g(t) + \boldsymbol{\beta}'\boldsymbol{x}}\right)^{-1/\rho}.$$

Mao and Wang (2010) refers to this class of latency submodels as the generalized proportional odds models.

For a prespecified ρ, if a nonparametric estimate of $g(t)$ is desired, the likelihood function (2.16) can be maximized within a class of nondecreasing step functions on $[0, \infty)$ with $g(0) = -\infty$ and with jumps only at the observed failure times $\tau_1, \ldots, \tau_k$. It is convenient to reparametrize the step function $g(t)$ as $\exp[g(t)] = \sum_{j:\tau_j \leq t} \exp(\alpha_j)$. Let $j_i = \max\{j : \tau_j \leq t_i\}$. Then (2.16) can be written as

$$\ell_2(\boldsymbol{\beta}, \boldsymbol{\alpha}, \rho) = \sum_{i=1, w_i^{(k)} > 0}^{n} w_i^{(k)} \left\{ \delta_i \left[\alpha_{j_i} - \log\left(\rho \sum_{j=1}^{j_i - 1} e^{\alpha_j} + e^{-\boldsymbol{\beta}'\boldsymbol{x}_i} \right) \right] \right.$$
$$\left. - \frac{1}{\rho} \left[\boldsymbol{\beta}'\boldsymbol{x}_i + \log\left(\rho \sum_{j=1}^{j_i} e^{\alpha_j} + e^{-\boldsymbol{\beta}'\boldsymbol{x}_i} \right) \right] \right\}. \tag{3.21}$$

A Newton-Raphson method may be used to maximize this function to update $\boldsymbol{\beta}$ and $\boldsymbol{\alpha}$.

Since k is in the same order as n and is potentially large, the high-dimensional $\boldsymbol{\alpha}$ may cause some numerical issues when directly apply the Newton-Raphson method to maximize (3.21). To overcome the issues, Mao and Wang (2010) suggested to use the minimization-maximization algorithm (Hunter and Lange, 2002). The minimization-maximization algorithm requires the construction for a simpler surrogate function than (3.21) to be maximized and is thus more computationally stable. However, it is not clear whether or not the surrogate function used in their work could be extended to semiparametric transformation models that are not the generalized proportional odds models.

The estimation method above requires a prespecified value for ρ. Unlike the parametric cases discussed in Section 2.5.2, there is little work on the inference of ρ in this semiparametric transformation model. Mao and Wang (2010) selected a grid of candidate values for ρ and chose the one that maximizes the profile likelihood function. Interval estimation of ρ and tests for a specific value of ρ for model selection may require a large sample size, and details remain to be worked out.

3.2.5 Semiparametric Incidence Submodel

Above semiparametric cure models assume a nonparametric baseline distribution for the latency submodels but leave the parametric logistic incidence submodels intact. To relax the parametric assumption in the incidence submodel, Li et al. (2020) suggested a more flexible support vector machine (SVM) approach to obtain $\pi(\boldsymbol{z}_i)$ in the M-step of the EM algorithm. That is, instead of maximizing (2.15) based on a parametric link such as the logit link or the complementary log-log link, an SVM method is considered here to update $\pi(\boldsymbol{z}_i)$.

If y_i's are all available, the SVM aims to build an optimal classification rule $g(\boldsymbol{z}) = \sum_{i=1}^n a_i(2y_i - 1)K(\boldsymbol{z}, \boldsymbol{z}_i) + b$ where $K(\cdot, \cdot)$ is a kernel function, a_i's and b are the solution of the following optimization problem:

$$\max_{a_1,\ldots,a_n,b} -\frac{1}{2}\sum_{i,j=1}^n a_i a_j (2y_i - 1)(2y_j - 1)K(\boldsymbol{z}_i, \boldsymbol{z}_j) + \sum_{i=1}^n a_i \tag{3.22}$$

subject to $\sum_{i=1}^n a_i(2y_i - 1) = 0$ and $0 \le a_i \le c$, $i = 1, 2, \ldots, n$, and c is a tuning parameter that controls the degree of the model overfitting. The kernel function $K(\cdot, \cdot)$ should be a symmetric positive (semi-) definite function. An example of this kernel function is the radial basis function $K(\boldsymbol{z}, \mathbf{v}) = e^{-\sigma(\boldsymbol{z}-\mathbf{v})'(\boldsymbol{z}-\mathbf{v})}$ where σ is the kernel width. Both c and σ can be estimated using the cross-validation method (Chang and Lin, 2011). The sequential minimal optimization method (Platt, 1998) can be used to solve (3.22). With the estimated function $g(\boldsymbol{z})$, an update of $\pi(\boldsymbol{z}_i)$ can be obtained using

the Platt scaling method (Platt et al., 1999). Let N_+ and N_- be the numbers of uncured and cured subjects in the data respectively, $\tilde{y}_i = \frac{N_+ + 1}{N_+ + 2}$ if $y_i = 1$ and $\tilde{y}_i = \frac{1}{N_- + 2}$ if $y_i = 0$, and $p_i = g(\boldsymbol{z}_i)$. The Platt scaling method estimates $\pi(\boldsymbol{z}_i)$ as $[1 + \exp(Ap_i + B)]^{-1}$ where A and B minimize

$$-\sum_{i=1}^{n} \{(1 - \tilde{y}_i)(Ap_i + B) - \log[1 + \exp(Ap_i + B)]\}. \tag{3.23}$$

Since y_i's are usually unknown for subjects with $\delta_i = 0$, a multiple imputation method can be used to obtain the values of y_i's with $\delta_i = 0$ based on the current $w_i^{(r)}$. That is, multiple imputed values of y_i from a Bernoulli distribution with probability $w_i^{(r)}$ are obtained. Then the rth update $\pi^{(r)}(\boldsymbol{z}_i)$ is obtained as the average of the estimates of $\pi(\boldsymbol{z}_i)$ from the multiple imputed samples, and the M-step is completed. The E-step and the M-step iterate until the EM algorithm converges.

It is recommended that covariates $\boldsymbol{x}_i$ and $\boldsymbol{z}_i$ be normalized when using the SVM method, that is, for the jth covariate x_{ij} in $\boldsymbol{x}_i$ or z_{ij} in $\boldsymbol{z}_i$, use $\frac{x_{ij} - \min_i(x_{ij})}{\max_i(x_{ij}) - \min_i(x_{ij})}$ instead of x_{ij} and use $\frac{z_{ij} - \min_i(z_{ij})}{\max_i(z_{ij}) - \min_i(z_{ij})}$ instead of z_{ij} in the model fitting.

As other semiparametric cure models, directly estimating the standard errors of estimates from the EM algorithm can be difficult. The bootstrap method should be employed to obtain the standard errors.

3.2.6 Semiparametric Spline-Based Cure Models

It is also possible to relax parametric assumptions in both parts of the semiparametric mixture cure models by using additive effects of the covariates in the two submodels. This idea was first discussed in Peng (2003b) by employing the generalized additive model for the incidence part provided in R function `gam` and additive effects in the Cox PH latency submodel provided in R function `coxph`. This method can be further extended to include interactions between two and more covariates in the two submodels (Wang et al., 2012a). Specifically, if $\boldsymbol{x} = (x_1, \ldots, x_{p_x})'$ and $\boldsymbol{z} = (z_1, \ldots, z_{p_z})'$, then the incidence part is modeled by

$$\pi(\boldsymbol{z}) = \frac{e^{\zeta(\boldsymbol{z})}}{1 + e^{\zeta(\boldsymbol{z})}}, \tag{3.24}$$

and the latency part is modeled by

$$h_u(t|\boldsymbol{x}) = e^{\eta(t, \boldsymbol{x})}. \tag{3.25}$$

where

$$\begin{aligned}\zeta(\boldsymbol{z}) =& \zeta_0 + \sum_{i=1}^{p_z} \zeta_i(z_i) + \sum_{ij} \zeta_{ij}(z_i, z_j) + \cdots \\ &+ \zeta_{1\cdots p_z}(z_1, \ldots, z_{p_z}) = \psi_\zeta(\boldsymbol{z})'\boldsymbol{\gamma}, \end{aligned} \tag{3.26}$$

$$\begin{aligned}\eta(t, \boldsymbol{x}) =& \eta_0 + \eta_t(t) + \sum_{i=1}^{p_x} \eta_i(x_i) + \sum_{ij} \eta_{ij}(x_i, x_j) + \cdots \\ &+ \eta_{1\cdots p_x}(x_1, \ldots, x_{p_x}) = \psi_\eta(t, \boldsymbol{x})'\boldsymbol{\beta}, \end{aligned} \tag{3.27}$$

and ψ_ζ and ψ_η are chosen spline basis functions. It can be seen that this approach uses splines, instead of specific linear or nonlinear effects, to model the main effects, two-way interaction effects, and high order interaction effects of x_i's in the latency part and similar effects of z_i's in the incidence part. The flexibility of splines reduces the dependence of the covariate effects on any particular parametric form. To reduce the dimensionality of $\boldsymbol{\beta}$ and $\boldsymbol{\gamma}$, the time–covariate interaction in the latency part, and higher-order interactions than two-way interactions in $\boldsymbol{x}$ and $\boldsymbol{z}$ may be excluded.

To estimate $\boldsymbol{\beta}$ and $\boldsymbol{\gamma}$, a penalized version of the EM algorithm in Section 2.3.2 can be used. That is, the likelihoods in (2.15) and (2.16) are replaced with their penalized versions:

$$\ell_1^P(\boldsymbol{\gamma}) = -\frac{1}{n}\ell_1(\boldsymbol{\gamma}) + \frac{\lambda_1}{2}J_1(\zeta), \tag{3.28}$$

$$\ell_2^P(\boldsymbol{\beta}, \boldsymbol{\alpha}) = -\frac{1}{n}\ell_2(\boldsymbol{\beta}, \boldsymbol{\alpha}) + \frac{\lambda_2}{2}J_2(\eta), \tag{3.29}$$

where $J_1(\zeta) = \sum_j \theta_{1j}^{-1}\|P_j(\zeta)\|^2$ is the roughness penalty with additional smoothing parameters θ_{1j} measuring the smoothness of ζ, and the smoothing parameter $\lambda_1(> 0)$ controls the tradeoff between bias and variance. Similarly $J_2(\eta) = \sum_j \theta_{2j}^{-1}\|P_j(\eta)\|^2$ is the roughness penalty with additional smoothing parameters θ_{2j} measuring the smoothness of η, and the smoothing parameter $\lambda_2(> 0)$ controls the tradeoff. Cubic and tensor cubic smoothing splines can be used in (3.26) and (3.27). The number of knots is suggested to be $10n^{2/9}$. The Newton-Raphson method can be used to minimize $\ell_1^P(\boldsymbol{\gamma})$ and $\ell_2^P(\boldsymbol{\beta}, \boldsymbol{\alpha})$, and the cross-validation scores are used in selecting smoothing parameters. Once $\boldsymbol{\beta}$ and $\boldsymbol{\gamma}$ are updated in the M-step, the E-step can proceed and the two steps iterate until the EM algorithm converges. The standard errors of estimated $\boldsymbol{\beta}$ and $\boldsymbol{\gamma}$ can be obtained using Louis (1982) method.

3.3 Nonparametric Mixture Cure Models

The semiparametric methods discussed in previous sections focus on semiparametric estimation in the latency part of the mixture cure models. The incidence part and the latency part beyond the baseline distribution are still parametrically specified. That is, the covariate effects on cure rate is modeled by specific link functions such as the logit link $\pi(\boldsymbol{z}) = \exp(\boldsymbol{\beta}'\boldsymbol{z})/(1+\exp(\boldsymbol{\beta}'\boldsymbol{z}))$ or the complementary log-log link $\pi(\boldsymbol{z}) = \exp(-\exp(\boldsymbol{\beta}'\boldsymbol{z}))$, and the covariate effects in the latency part is always in the form of $\exp(\boldsymbol{\beta}'\boldsymbol{x})$. When the underlying covariate effects in the cure model cannot be well approximated by the parametric assumptions in any of the two parts, applying the semiparametric models can still lead to biased estimates. In this section, we introduce nonparametric methods to estimate the effects of covariates in the latency part that do not require any parametric assumptions. We will also introduce methods to nonparametrically estimate covariate effects in the latency part.

3.3.1 Nonparametric Incidence Submodels

3.3.1.1 Kaplan-Meier Estimator

We start with a nonparametric estimation of cure rate for a single sample without covariates. Let $\hat{S}^{\mathrm{KM}}(t)$ be the Kaplan-Meier survival estimate of $S(t)$ based on data $\{(t_i, \delta_i), i = 1, \ldots, n\}$, and $t_{(n)} = \max\{t_i\colon i = 1, \ldots, n\}$. Since $1 - \pi = \lim_{t\to\infty} S(t)$ in (2.1), the simplest cure rate estimate is the lowest value of $\hat{S}^{\mathrm{KM}}(t)$. That is, a nonparametric estimate of π is

$$\hat{\pi} = 1 - \hat{S}^{\mathrm{KM}}(t_{(n)}). \tag{3.30}$$

It can be shown that $\hat{\pi}$ in (3.30) is a consistent estimator of π and is asymptotically normal with the variance that can be estimated by (Maller and Zhou, 1992)

$$\hat{\pi}^2 \sum_{j=1}^{k} \frac{d_j}{n_j(n_j - d_j)},$$

where k and d_j are defined as in Section 3.2 and $n_j = \sum_{i=1}^{n} Y_i(\tau_j)$ is the number of subjects at risk at τ_j. Let $\tau_u = \inf\{t\colon S_u(t) = 0\}$ and $\tau_G = \inf\{t\colon G(t) = 0\}$ where $G(t)$ is the survival function of the censoring distribution. That is, τ_u and τ_G are the right extremes of the supports of the latency distribution and the censoring distribution respectively. The main condition for the asymptotic properties is that $\tau_u \le \tau_G$, which implies that a sufficient follow-up is required for a nonparametric cure rate estimation.

This method can be generalized to multiple sample cases (Laska and Meisner, 1992; Tsodikov, 2001). For the two-sample leukemia data set with the

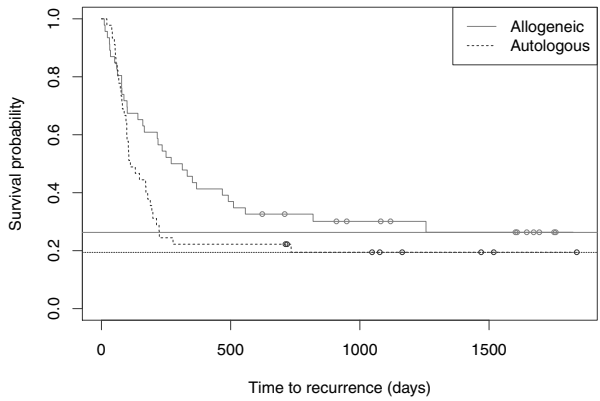

FIGURE 3.1
Nonparametric cure rate estimates indicated by the horizontal lines at the heights of the right tails of the Kaplan-Meier survival curves for the two groups of leukemia patients with marrow transplants.

Kaplan-Meier survival estimates in Figure 2.4, the corresponding nonparametric estimates of the cure rates $1 - \hat{\pi}$ from the two groups are marked in Figure 3.1.

3.3.1.2 Generalized Kaplan-Meier Estimator

The above method can be viewed as a method to model the incidence part with a single categorical variable z. The method however cannot handle continuous covariates in data. To model the effect of a continuous covariate z nonparametrically in the incidence part, the generalized Kaplan-Meier estimator of Beran (1981) can be considered. For survival data $(t_i, \delta_i, z_i), i = 1, \ldots, n$, the generalized Kaplan-Meier estimator for $S(t|z)$ is given by

$$\hat{S}^{\mathrm{GKM}}(t|z) = I(t \leq t_{(n)}) \prod_{j=1}^{n} \left[\frac{1 - \sum_{r=1}^{n} I(t_r \leq t_j) B_r(z)}{1 - \sum_{r=1}^{n} I(t_r < t_j) B_r(z)} \right]^{I(t_j \leq t; \delta_j = 1)}, \quad (3.31)$$

where $B_j(z)$ is a proper weight function satisfying $B_j(z) \geq 0$ and $\sum_{j=1}^{n} B_j(z) = 1$. As a generalization of the prominent Kaplan-Meier survival estimator, the generalized Kaplan-Meier estimator (3.31) has been widely used to estimate the survival function with covariates. The weight function $B_j(z)$ can be the Nadaraya-Watson weight (Nadaraya, 1964; Watson, 1964) function

$$B_j(z) = \frac{K_h(z_j - z)}{\sum_{r=1}^{n} K_h(z_r - z)}, \quad (3.32)$$

where $K_h(\cdot)$ is a kernel density function as defined in Section 3.2.2.3. If the covariate z is continuous, the following Gasser-Müller weight (Gasser and Müller,

1979) function can also be used in (3.31),

$$B_j(z) = \int_{s_{j-1}}^{s_j} K_h(z-u)du,$$

where $s_0 = 0, s_j = \frac{1}{2}(z_j + z_{j+1}), j = 1, \ldots, n-1$ and $s_n = 1$.

Following the same idea in Section 3.3.1.1, if a cure fraction is present in data, a nonparametric estimate of $\pi(z)$ can be obtained from (3.31) by (Xu and Peng, 2014)

$$\hat{\pi}(z) = \hat{S}^{\mathrm{GKM}}(t^1_{(n)}|z), \tag{3.33}$$

where $t^1_{(n)} = \max_{i:\,\delta_i=1}(t_i)$ is the largest uncensored failure time. Under mild conditions, it can be shown that the estimator (3.33) is a consistent estimator of $\pi(z)$, where $0 < \pi(z) < 1$ for any interior point z, and the estimator is asymptotically normal with the variance estimated by $\hat{\pi}^2(z)\hat{\sigma}^2(z)$, where

$$\hat{\sigma}^2(z) = \int K^2(t)dt \sum_{j=1}^{n} \frac{\delta_j B_j(z)}{\hat{H}(t_j|z)\hat{H}(t_j-|z)},$$

and $H(t|z) = P[\min(T, C) > t|z]$ and

$$\hat{H}(t|z) = 1 - \sum_{j=1}^{n} I(t_j \le t)B_j(z).$$

The conditions required for the asymptotic properties include those typically for smooth nonparametric estimators, kernel function, Gasser-Müller weight function, and the bandwidth h. The most prominent condition is that for any z

$$\tau_u < \tau_G(z) \tag{3.34}$$

with probability 1, where $\tau_u = \sup_{z\in D} \tau_{S_u}(z)$, $\tau_{S_u}(z) = \inf\{t\colon S_u(t|z) = 0\}$, D is the support of z, and $\tau_G(z) = \inf\{t\colon G(t|z) = 0\}$. This condition guarantees a sufficient follow-up so that censored subjects beyond the largest observed failure time are most likely to be cured and hence our estimator does not overestimate the true cure rate. Amico and Van Keilegom (2018) suggested that this condition be further relaxed into

$$\tau_u(z) < \tau_G(z) \tag{3.35}$$

where $\tau_u(z) = \inf\{t\colon S_u(t|z) = 0\}$. In practice, we find that as long as the censoring distribution $G(t|z)$ has a heavier tail than $S_u(t|z)$, $\hat{\pi}(z)$ in (3.33) will tend to have smaller biases regardless of the value of $\tau_{S_u(z)}$.

The convergence rate of (3.33) to $\pi(z)$ is related to the convergence rate of the convergence of generalized Kaplan-Meier survival estimator to $S(t)$, which is generally slower than $n^{-1/2}$ compared to the Kaplan-Meier survival estimate

because it only uses a portion of data. The actual rate of convergence depends on how fast h goes to 0 as well as on the smoothness of $S(t)$. A decrease in the bandwidth would reduce the bias but, on the other hand, increase the variance of the estimate, and vice versa. An optimal bandwidth that minimizes the asymptotic mean square error is given by (López-Cheda et al., 2017a)

$$h = \left(\frac{c_K \tilde{\sigma}^2(z)}{d_K^2 \mu^2(z)} \right)^{1/5} n^{-1/5} \tag{3.36}$$

where c_K, $\tilde{\sigma}^2(z)$, d_K^2, and $\mu(z)$ depend on the kernel function $K_h(\cdot)$, the density of the covariate z, and $H(t|z)$. Thus, (3.36) is not very useful to determine the value of h in practice. A more practical method to determine the bandwidth is to use a bootstrap method. Details can be found in López-Cheda et al. (2017a).

When there is no covariate, $B_j(\cdot) = 1/n$ for $j = 1, \ldots, n$ by (3.32) and hence the estimator (3.33) reduces to the single sample cure rate estimator (3.30).

The proposed method can be extended to multiple covariates. If $\boldsymbol{z} = (z_1, \ldots, z_{p_z})'$, one may use the product kernel (Simonoff, 2012), i.e., $K(\boldsymbol{z}) = \prod_{i=1}^{p_z} K(z_i)$, in (3.33) to estimate $\pi(\boldsymbol{z})$.

3.3.2 Nonparametric Latency Submodels

If the same covariate affects both the latency and the incidence parts of the mixture cure model as follows

$$S(t|x) = \pi(x) S_u(t|x) + 1 - \pi(x)$$

then given nonparametric estimates of $S(t|x)$ and $\pi(x)$, such as (3.31) and (3.33), a nonparametric estimate of the latency part can be obtained as follows (López-Cheda et al., 2017b):

$$\hat{S}_u(t|x) = \frac{\hat{S}^{\mathrm{GKM}}(t|x) - 1 + \hat{\pi}(x)}{\hat{\pi}(x)} \tag{3.37}$$

One issue with this estimator is the bandwidth selection. The optimal bandwidths for $\hat{S}^{\mathrm{GKM}}(t|x)$ and $\hat{\pi}(x)$ can be different. However, different bandwidths may not yield a proper survival function estimate for $S_u(t|x)$ since the limit of $\hat{S}^{\mathrm{GKM}}(t|x)$ as $t \to \infty$ needs not to be $\hat{\pi}(x)$ with different bandwidths. Therefore the same bandwidth is recommended in (3.37). Under similar conditions and a suitable bandwidth h, it can be shown that $\hat{S}_u(t|x)$ is a consistent estimate of $S_u(t|x)$ and is asymptotically normal with variance $V(t,z)c_K$, where $V(t,z)$ depends on $\pi(x)$, $S(t|x)$, $H(t|x)$, and the density of x. The bandwidth can be selected using the bootstrap method. See details in López-Cheda et al. (2017b).

One obvious limitation of the estimator (3.37) is that it requires the two parts of the mixture cure model share the same covariate. If different covariates are required in the two parts, the method will not be appropriate to estimate $S_u(t|x)$ directly. One could allow all covariates to appear in both parts so that (3.37) can be used. However, the estimator may produce a spurious relationship between some covariates and the two submodels of the mixture cure model, and the estimation may not be efficient.

Even though the generalized Kaplan-Meier survival estimator can be used with more than one covariate by choosing appropriate kernel functions that can work with multiple covariates, the application of the method in practice can be challenging with more than one covariate.

3.4 Semiparametric Non-Mixture Cure Models

3.4.1 Semiparametric PHC Model

In addition to the parametric approaches to estimate non-mixture cure models discussed in Section 2.4, there are also semiparametric approaches to estimate the non-mixture cure models if a parametric distributional assumption is not desirable. For the PHC model in (2.18), given its similarity to the PH model, a similar semiparametric estimation method to the PH model can be used to estimate the PHC model (Tsodikov, 1998b). Particularly, estimating $\boldsymbol{\gamma}$ in (2.18) can be obtained using the partial log-likelihood function:

$$\log \prod_{j=1}^{k} \frac{\exp(\boldsymbol{\gamma}'\boldsymbol{s}_j)}{\left\{\sum_{i\in R_j} \exp(\boldsymbol{\gamma}'\boldsymbol{z}_i)\right\}^{d_j}},$$

following the notations in Section 3.2, where $\boldsymbol{s}_j = \sum_{i:t_i=\tau_j} \delta_i \boldsymbol{z}_i$. However, if γ_0 or $F^H(t)$ is of interest, the partial likelihood function is not very useful, and a usual nonparametric baseline survival function estimation method for Cox's PH model can be used to estimate $\exp[-e^{\gamma_0}F^H(t)]$. For example, given $\hat{\boldsymbol{\gamma}}$, the estimate of $\boldsymbol{\gamma}$ from the partial log-likelihood function, a Nelson-Aalen estimator of the baseline survival function is given by

$$\widehat{S}_0(t) = \exp\left(-\sum_{j:\tau_j<t} \frac{d_j}{\sum_{i\in R_j} \exp(\hat{\boldsymbol{\gamma}}'\boldsymbol{z}_i)}\right).$$

Then following the idea in Section 3.3.1.1, γ_0 can be estimated by $\hat{\gamma}_0 = \log[-\log(\widehat{S}_0(\tau_k))]$, and $F^H(t)$ can be estimated by $\widehat{F^H(t)} = -\log[\widehat{S}_0(t)^{\exp(-\hat{\gamma}_0)}]$ (Tsodikov et al., 2003), which essentially implies that $\widehat{F^H(\tau_k)} \equiv 1$ and any subjects with censored times greater than τ_k is deemed

cured. Assuming the censored subjects with survival times greater than the largest uncensored time as cured is often necessary in semiparametric estimation methods to ensure the PHC model is identifiable, similar to what is discussed in Section 3.2.1.1 for semiparametric mixture cure models.

Other approaches, such as the marginal likelihood approach of the PH model (Kalbfleisch and Prentice, 2002), can also be extended to estimate the PHC model semiparametrically (Tsodikov, 1998b). Asymptotic properties of the estimates and a comparison with a parametric PHC model are discussed in detail in Tsodikov (1998a, 2001).

3.4.2 General Non-Mixture Cure Models

The general non-mixture cure models based on Box-Cox transformation in Section 2.4.4 can be estimated semiparametrically as well. Let $g(\cdot|\lambda)$ denote the transformation and the corresponding survival function of the cure model is $S(t|\boldsymbol{z}) = g[e^{\gamma_0+\boldsymbol{\gamma}'\boldsymbol{z}}F^H(t)]$. For example, model (2.26) corresponds to

$$g(x|\lambda) = \begin{cases} (1+\lambda x)^{-1/\lambda} & \lambda > 0 \\ e^{-x} & \lambda = 0. \end{cases}$$

Without making any parametric assumption for $F^H(t)$, we consider a nonparametric maximum likelihood estimation approach, where $F^H(t)$ is allowed to be a right-continuous function. Let $\boldsymbol{\alpha}' = (\alpha_1, \ldots, \alpha_n)$, where α_i be the jump size of $F^H(t)$ at t_i. The observed likelihood function is

$$\prod_{i=1}^{n} \left\{ \left(-g'[e^{\gamma_0+\boldsymbol{\gamma}'\boldsymbol{z}_i}F^H(t_i)]e^{\gamma_0+\boldsymbol{\gamma}'\boldsymbol{z}_i}\alpha_i\right)^{\delta_i} \left(g[e^{\gamma_0+\boldsymbol{\gamma}'\boldsymbol{z}_i}F^H(t_i)]\right)^{1-\delta_i} \right\}$$

subject to $\sum_i \delta_i\alpha_i = 1$, where $F^H(t_i) = \sum_{j:t_j\le t_i} \delta_j\alpha_j$. Suppose that t_i's are arranged from smallest to largest and $m = \sum_i I(t_i \le \tau_k)$. Because of the identifiability requirement as discussed earlier, it is easy to see that $\alpha_i \equiv 0$ for $i > m$. For the remaining α_i, $i = 1, \ldots, m$, γ_0 and $\boldsymbol{\gamma}$, it can be shown (Zeng et al., 2006) that their constrained maximum likelihood estimates can

be obtained from the following equations:

$$
\begin{aligned}
0 = & \sum_{i=1}^{m} \frac{1}{\alpha_i} \frac{\partial \alpha_i}{\partial(\gamma_0, \boldsymbol{\gamma}')'} + \sum_{i=1}^{n} \delta_i \boldsymbol{z}_i - C \sum_{i=1}^{m} \frac{\partial \alpha_i}{\partial(\gamma_0, \boldsymbol{\gamma}')'} \\
& + \sum_{i=m+1}^{n} \frac{g'[e^{\gamma_0+\boldsymbol{\gamma}'\boldsymbol{z}_i}]}{g[e^{\gamma_0+\boldsymbol{\gamma}'\boldsymbol{z}_i}]} e^{\gamma_0+\boldsymbol{\gamma}'\boldsymbol{z}_i} \boldsymbol{z}_i \\
& + \sum_{i=1}^{m} \frac{g''[e^{\gamma_0+\boldsymbol{\gamma}'\boldsymbol{z}_i} F^H(t_i)]}{g'[e^{\gamma_0+\boldsymbol{\gamma}'\boldsymbol{z}_i} F^H(t_i)]} \left[\boldsymbol{z}_i F^H(t_i) + \frac{\partial F^H(t_i)}{\partial(\gamma_0, \boldsymbol{\gamma}')'} \right] e^{\gamma_0+\boldsymbol{\gamma}'\boldsymbol{z}_i} \\
& + \sum_{i=1}^{m} \sum_{t_i < t_j \le t_{i+1}} \frac{g'[e^{\gamma_0+\boldsymbol{\gamma}'\boldsymbol{z}_j} F^H(t_i)]}{g[e^{\gamma_0+\boldsymbol{\gamma}'\boldsymbol{z}_j} F^H(t_i)]} \left[\boldsymbol{z}_j F^H(t_i) + \frac{\partial F^H(t_i)}{\partial(\gamma_0, \boldsymbol{\gamma}')'} \right] e^{\gamma_0+\boldsymbol{\gamma}'\boldsymbol{z}_j} \\
0 = & \sum_{i=1}^{m} \frac{1}{\alpha_i} \frac{\partial \alpha_i}{\partial \alpha} - C \sum_{i=1}^{m} \frac{\partial \alpha_i}{\partial \alpha} + \sum_{i=1}^{m} \frac{g''[e^{\gamma_0+\boldsymbol{\gamma}'\boldsymbol{z}_i} F^H(t_i)]}{g'[e^{\gamma_0+\boldsymbol{\gamma}'\boldsymbol{z}_i} F^H(t_i)]} \frac{\partial F^H(t_i)}{\partial \alpha} e^{\gamma_0+\boldsymbol{\gamma}'\boldsymbol{z}_i} \\
& + \sum_{i=1}^{m} \sum_{t_i < t_j \le t_{i+1}} \frac{g'[e^{\gamma_0+\boldsymbol{\gamma}'\boldsymbol{z}_j} F^H(t_i)]}{g[e^{\gamma_0+\boldsymbol{\gamma}'\boldsymbol{z}_j} F^H(t_i)]} \frac{\partial F^H(t_i)}{\partial \alpha} e^{\gamma_0+\boldsymbol{\gamma}'\boldsymbol{z}_j} \\
0 = & \sum_{i=1}^{m} \alpha_i - 1.
\end{aligned}
$$

where $\alpha \equiv \alpha_m$ and C is the Lagrange multiplier. The derivative of α_i with respect to γ_0, $\boldsymbol{\gamma}$ and α can be obtain from a recursive formula between α_i and α_{i+1}:

$$
\begin{aligned}
\frac{1}{\alpha_i} = \frac{1}{\alpha_{i+1}} - & \frac{g''[e^{\gamma_0+\boldsymbol{\gamma}'\boldsymbol{z}_i} F^H(t_i)] e^{\gamma_0+\boldsymbol{\gamma}'\boldsymbol{z}_i}}{g'[e^{\gamma_0+\boldsymbol{\gamma}'\boldsymbol{z}_i} F^H(t_i)]} \\
& - \sum_{t_i < t_j \le t_{i+1}} \frac{g'[e^{\gamma_0+\boldsymbol{\gamma}'\boldsymbol{z}_j} F^H(t_i)] e^{\gamma_0+\boldsymbol{\gamma}'\boldsymbol{z}_j}}{g[e^{\gamma_0+\boldsymbol{\gamma}'\boldsymbol{z}_j} F^H(t_i)]}.
\end{aligned}
$$

The semiparametric method above is for a fixed value of λ in $g(\cdot|\lambda)$. If one is interested in estimating the value of λ using data, one has to ensure that λ is identifiable. For model (2.26), it can be shown that if $\boldsymbol{\gamma}'\boldsymbol{z} \neq 0$, then γ_0, $F^H(t)$ and λ are identifiable. In this case, λ_0 can be jointly estimated with other parameters in the model. Or one can use the AIC or BIC to select λ from preselected candidate values.

Another generalization of the semiparametric PHC model is the nonlinear transformation model (NTM) (Tsodikov et al., 2003), in which the survival function of T^* is given by

$$S(t|\boldsymbol{z}) = g[S^H(t)|\boldsymbol{z}], \tag{3.38}$$

where $g(\cdot|\boldsymbol{z})$ some parametrically specified cumulative distribution function with support on [0, 1]. The model can also be defined in survival function as

$$S(t|\boldsymbol{z}) = E[S^H(t)^M|\boldsymbol{z}], \tag{3.39}$$

where M is a distribution that has a probability mass at 0 and the expectation is with respect to M. If $g(x|\boldsymbol{z}) = \exp(-e^{\gamma_0+\boldsymbol{\gamma}'\boldsymbol{z}}(1-x))$ or if M follows a Poisson distribution with mean $e^{\gamma_0+\boldsymbol{\gamma}'\boldsymbol{z}}$, the above models reduce to the PHC model (2.18). If $M = \exp(\boldsymbol{\beta}'\boldsymbol{x})v$ and v follows a Poisson distribution with mean $e^{\gamma_0+\boldsymbol{\gamma}'\boldsymbol{z}}$, then the above model reduces to the PHPH cure model (2.19). If $M = \exp(\boldsymbol{\beta}'\boldsymbol{x})v$ and v follows a Bernoulli distribution with mean $1/[1+\exp(-\boldsymbol{\gamma}'\boldsymbol{z})]$, the above model reduces the PH mixture cure model (2.5). A necessary and sufficient condition for the two forms (3.38) and (3.39) to be equivalent is that the function $\psi(t) = g(e^{-t})$ has $(-1)^n\psi^{(n)}(t) \geq 0$ for any n.

Given observed data $(t_i, \delta_i, \boldsymbol{x}_i, \boldsymbol{z}_i)$, $i = 1, \ldots, n$, the likelihood function can be written as

$$\ell = \sum_{i=1}^{n} \left\{\delta_i \log\left(g[S^H(t_i-)] - g[S^H(t_i)]\right) + (1-\delta_i)\log g[S^H(t_i)]\right\}.$$

A semiparametric estimation method often maximizes the likelihood function for a step function $S^H(t)$ with steps at the uncensored times. It is also necessary to require that $\lim_{t\to\infty} \widehat{S^H}(t) = 0$ for $\widehat{S^H}(t)$ as an estimate of $S^H(t)$ and there is no intercept term in $\boldsymbol{\beta}$. The unknown parameters in this model include $\boldsymbol{\gamma}$ and the steps in $S^H(t)$ (and $\boldsymbol{\beta}$ if covariates $\boldsymbol{x}$ are considered as in the PHPH model).

The likelihood function can be maximized using the Newton-Raphson method or the profile likelihood method. However, these methods may fail sometimes due to the high dimensional steps in $S^H(t)$. Alternative approaches include a restricted nonparametric maximum likelihood estimation method (RNPMLE) (Tsodikov, 2002) and a quasi-EM (QEM) algorithm (Tsodikov, 2003) can be considered. In the RNPMLE method, the estimates of the unknown parameters are obtained from the following estimating functions

$$\begin{aligned} 0 =& \frac{\partial \ell}{\partial \boldsymbol{\gamma}} \\ 0 =& \frac{\partial \ell}{\partial F^H(t_j)} = g_i(F^H(t_{j-1}), F^H(t_j), F^H(t_{j+1})), \quad j = 1, \ldots, n. \end{aligned}$$

The last equation can be solved recurrently given $F^H(t_0) = 0$ and $F^H(t_n) = 1$.

The QEM algorithm is similar to the EM algorithm except that it uses a distribution-free E-step in the EM algorithm where M in (3.39) is treated as the latent variable. The distribution-free method for the E-step is to update the conditional expectation of M by

$$E[M(\cdot)|t_i, \delta_i] = \widetilde{M}[S^H(t_i)|\cdot, \delta_i],$$

where

$$\widetilde{M}[x|\cdot, \delta] = \delta + \frac{g^{(\delta+1)}(x|\cdot)}{g^{(\delta)}(x|\cdot)}.$$

For the PHC model (2.18), $\widetilde{M}[x|\cdot,\delta] = \delta + xe^{\gamma_0+\boldsymbol{\gamma}'\boldsymbol{z}}$, and for the PHPH cure model (2.19), $\widetilde{M}[x|\cdot,\delta] = \delta e^{\boldsymbol{\beta}'\boldsymbol{x}} + x^{\exp(\boldsymbol{\beta}'\boldsymbol{x})}e^{\boldsymbol{\beta}'\boldsymbol{x}+\gamma_0+\boldsymbol{\gamma}'\boldsymbol{z}}$. The estimate of the step of the corresponding cumulative hazard function of $F^H(t)$ at the jth uncensored time is given by $d_j \Big/ \sum_{l\in R_j} \widetilde{M}[S^H(t_l)|\boldsymbol{\gamma},\delta_l,\boldsymbol{z}_l]$. Details of the QEM algorithm can be found in Tsodikov et al. (2003) and Tsodikov (2003).

3.5 Model Assessment

3.5.1 Residuals for Overall Model Fitting

Residuals are often used in statistical models to examine the overall model fitting. The generalized residuals considered in Section 2.5.3, however, are primarily used for parametric mixture cure models because of their dependence on the hazard functions. Their use for semiparametric mixture cure models is limited because the estimated hazard function is usually not directly available in most semiparametric mixture cure models.

For semiparametric mixture cure models, some residual-based methods for standard survival models may be used to assess the fit of the model. For example, based on (3.16), the martingale residual for model (2.5) can be defined as

$$M_i = \delta_i + \log\left[1 + e^{\boldsymbol{\gamma}'\boldsymbol{z}_i - H_{u0}(t_i)\exp(\boldsymbol{\beta}'\boldsymbol{x}_i)}\right] - \log[1 + e^{\boldsymbol{\gamma}'\boldsymbol{z}_i}]. \tag{3.40}$$

Let $\hat{M}_i$ be the estimated M_i when $\boldsymbol{\beta}$, $H_{u0}(t)$ and $\boldsymbol{\gamma}$ are replaced with their estimates. The martingale residual can be viewed as a difference between the observed number of events for an individual and the conditionally expected number given the fitted mixture cure model, follow-up time, and covariates. It shares some of the properties of the martingale residuals for classical survival models, such as $M_i \leq 1$, $E(M_i) = 0$, $\sum \hat{M}_i = 0$, $Cov(M_i, M_j) = 0$, and $Cov(\hat{M}_i, \hat{M}_j) < 0$. However, the martingale residual in (3.40) has a lower bound $\log[1-\pi(z_i)]$ while in the classical survival models the martingale residual does not have a lower bound (Peng and Taylor, 2017). The presence of the lower bound may limit the use of the martingale residuals to detect outliers. Numerical studies also show that the martingale residuals are insensitive to the covariate effects in the incidence part of the cure model.

With the martingale residuals M_i in (3.40), the Cox-Snell residuals can be defined by

$$r_i = \delta_i - M_i, \quad i = 1, \ldots, n, \tag{3.41}$$

for the mixture cure model (2.1). Under a standard survival model, if the model is correctly specified and is consistently estimated, then the Cox-Snell residuals can be viewed as a censored sample from a unit exponential distribution. However, under the mixture cure model (2.1), the Cox-Snell residuals

(r_i, δ_i), $i = 1, \ldots, n$ can no longer be viewed as a sample from a unit exponential distribution when the cure model is correctly specified. This can be seen from the fact that

$$P[-\log S(T|\boldsymbol{x}, \boldsymbol{z}) > t] = \begin{cases} e^{-t} & \text{if } t < -\log[1 - \pi(\boldsymbol{z})] \\ 0 & \text{if } t \geq -\log[1 - \pi(\boldsymbol{z})] \end{cases}.$$

It is a mixed type distribution with a unit exponential distribution as the continuous component between 0 and $-\log[1-\pi(\boldsymbol{z})]$ and a probability mass $1-\pi(\boldsymbol{z})$ at $-\log[1-\pi(\boldsymbol{z})]$ as the discrete component. Despite this fact, the usual approach of checking the estimated cumulative hazard rate of the (r_i, δ_i)'s against the cumulative hazard function of the unit exponential distribution is still appropriate because the residuals that are equal to $-\log[1 - \pi(\boldsymbol{z})]$ are always censored and thus the entire residuals still can be treated as a censored sample from the unit exponential distribution.

When the mixture cure model (2.1) is misspecified, the Cox-Snell residuals (3.41) will still follow a mixed type distribution. Suppose that the incidence submodel of the mixture cure model (2.1) is misspecified, but the latency submodel is still correct. Let $\pi^*(\boldsymbol{z})$ be the incorrect model for the uncure rate and $\pi(\boldsymbol{z})$ be the true model for the uncure rate. Then the survival function of the Cox-Snell residuals is

$$P[-\log S(T|\boldsymbol{x}, \boldsymbol{z}) > t] = \begin{cases} 1 - \frac{\pi(\boldsymbol{z})}{\pi^*(\boldsymbol{z})} + \frac{\pi(\boldsymbol{z})}{\pi^*(\boldsymbol{z})} e^{-t} & \text{if } t < -\log[1 - \pi^*(\boldsymbol{z})] \\ 0 & \text{if } t \geq -\log[1 - \pi^*(\boldsymbol{z})] \end{cases}.$$

It is a mixed type distribution with a non-unit exponential distribution as the continuous component between 0 and $-\log[1 - \pi^*(\boldsymbol{z})]$ and a probability mass $1 - \pi(\boldsymbol{z})$ at $-\log[1 - \pi^*(\boldsymbol{z})]$ as the discrete component. If the incidence part is correct but the latency part is misspecified, and the misspecified latency submodel is $S_u^*(t|\boldsymbol{x})$, the survival function of the Cox-Snell residuals is

$$\begin{cases} 1 - \pi(\boldsymbol{z}) + \pi(\boldsymbol{z}) S_u \left[S_u^{*-1} \left(\frac{e^{-t}}{\pi(\boldsymbol{z})} - \frac{1-\pi(\boldsymbol{z})}{\pi(\boldsymbol{z})} \right) \right] & \text{if } t < -\log[1 - \pi(\boldsymbol{z})] \\ 0 & \text{if } t \geq -\log[1 - \pi(\boldsymbol{z})] \end{cases}.$$

It is again a mixed type distribution with a distribution of a mixture cure model with an unknown distribution as the continuous component between 0 and $-\log[1 - \pi(\boldsymbol{z})]$ and a probability mass $1 - \pi(\boldsymbol{z})$ at $-\log[1 - \pi(\boldsymbol{z})]$ as the discrete component.

Similar to martingale residuals, numerical studies show that the Cox-Snell residuals in the mixture cure models are sensitive to the choice of the model assumption in the latency submodel but less sensitive to the misspecification in the incidence submodel.

3.5.2 Residuals for Latency Submodels

Residuals are also available for examining the fit of the latency submodel in the semiparametric mixture cure model. If the latency submodel is correctly

specified for $T|Y = 1$, the martingale residuals for the fitted latency submodel for uncured subjects can be defined as

$$M_i^{(u)} = \delta_i - y_i H_u(t_i|\boldsymbol{x}_i). \tag{3.42}$$

The estimated $\hat{M}_i^{(u)}$ can be obtained by replacing y_i with w_i in (2.14), and H_u with its estimate from the EM algorithm at convergence. For the PH mixture cure model (2.1) and (2.5), the estimated martingale residual for an uncured subject i is (Peng and Taylor, 2017)

$$\hat{M}_i^{(u)} = \delta_i - w_i \hat{H}_{u0}(t_i) \exp(\hat{\boldsymbol{\beta}}' \boldsymbol{x}_i). \tag{3.43}$$

This modified martingale residual shares similar properties as the martingale residual for the standard PH model. For example, if the baseline distribution is estimated by a Nelson-Aalen estimator (3.3), it can be shown that $\sum_{i=1}^n \hat{M}_i^{(u)} = 0$. Because of this condition, $\hat{M}_i^{(u)}$ and $\hat{M}_j^{(u)}$ will be slightly negatively correlated even though $Cov[M_i^{(u)}(t), M_j^{(u)}(t)] = 0$. The modified martingale residuals (3.43) may be used to examine the functional forms of covariates in x in the latency PH model in the same way as in the standard PH model (Therneau and Grambsch, 2000).

Similarly, a score process as a function of the modified martingale process can be defined as

$$U(\hat{\boldsymbol{\beta}}, t) = \sum_{i=1}^n \int_0^t [\boldsymbol{x}_i - \bar{\boldsymbol{x}}(u)]\, d\hat{M}_i^{(u)}(u) = \sum_{i=1}^n \boldsymbol{x}_i \hat{M}_i^{(u)}(t), \tag{3.44}$$

where $\bar{\boldsymbol{x}}(u) = \frac{\sum_{i=1}^n Y_i(u) w_i \boldsymbol{x}_i e^{\hat{\boldsymbol{\beta}}' \boldsymbol{x}_i}}{\sum_{i=1}^n Y_i(u) w_i e^{\hat{\boldsymbol{\beta}}' \boldsymbol{x}_i}}$. It is a special case of a cumulative sum $W_x(t, \boldsymbol{x}) = \sum_{i=1}^n g(\boldsymbol{x}_i) I(\boldsymbol{x}_i \leq \boldsymbol{x}) \hat{M}_i^{(u)}(t)$, where $g(\cdot)$ is a known smooth function. The cumulative sums from the Cox's PH model converge to a zero-mean Gaussian process if the PH model is correctly specified and consistently estimated, a property that can be used to check the functional form of a covariate, the exponential link function, and the PH assumption in the Cox's PH model. And $U(\hat{\boldsymbol{\beta}}, t)$ is asymptotically equivalent to a function of independent standard normal random variables, which facilitates a resampling approach to approximate the distribution of $U(\hat{\boldsymbol{\beta}}, t)$ (Lin et al., 1993). A similar property can be obtained for the latency PH submodel following the fact that w_i can be treated as an offset term in the latency PH submodel. Thus, the fit of the latency PH submodel can be examined based on $W_x(t, \boldsymbol{x})$ in a similar way as Cox's PH model without a cure fraction. For example, let $U_j(\hat{\boldsymbol{\beta}}, t)$ be the jth component of $U(\hat{\boldsymbol{\beta}}, t)$, $\mathscr{I}(\hat{\boldsymbol{\beta}})$ be minus the derivative matrix of $U(\hat{\boldsymbol{\beta}}, \infty)$, and $\mathscr{I}^{-1}(\hat{\boldsymbol{\beta}})_{jj}$ be the element of the inverse of the matrix at the jth row and jth column. A visual examination of $U_j(\hat{\boldsymbol{\beta}}, t)$ versus t against simulated zero-mean Gaussian processes will reveal a departure from the assumptions in the

latency PH submodel. The departure from the PH assumption for covariate $\boldsymbol{x}_j$ can be tested based on the Kolmogorov-type supremum test:

$$\sup_{t\in[0,\tau]} \|U_j(\hat{\boldsymbol{\beta}}, t)\| \quad \text{or} \quad \sup_{t\in[0,\tau]} \left\| \sqrt{\mathscr{I}^{-1}(\hat{\boldsymbol{\beta}})_{jj}} U_j(\hat{\boldsymbol{\beta}}, t) \right\|. \tag{3.45}$$

The p-value of the test can be obtained as the proportion of the values of (3.45) based on the simulated processes that are greater than the values of (3.45) based on the observed process.

A Schoenfeld residuals for the latency part can also be defined as

$$\delta_i[\boldsymbol{x}_i - \bar{\boldsymbol{x}}(t_i)], \tag{3.46}$$

which corresponds to the first part of $\int_0^t [\boldsymbol{x}_i - \bar{\boldsymbol{x}}(u)]\, d\hat{M}_i^{(u)}(u)$ in (3.44) evaluated at $t = \infty$. In comparison, the modified Schoenfeld residuals discussed in Section (2.5.3) is not appropriate for assessing the latency part because it depends on $h(t|\boldsymbol{x}, \boldsymbol{z})$. If $h(t|\boldsymbol{x}, \boldsymbol{z})$ is replaced with $h_{u0}(t)\exp(\boldsymbol{\beta}'\boldsymbol{x})$ under the PH mixture cure model, the residual in Section (2.5.3) is reduced to the modified Schoenfeld residual for the latency part in (3.46).

A Cox-Snell residual for the latency submodel for subjects with $u_i = 1$ can also be defined as follows (Peng and Taylor, 2017)

$$r_i^u = -\log S_u(t_i|\boldsymbol{x}_i). \tag{3.47}$$

To examine the distribution of r_i^u, the survival function of the distribution can be estimated using a weighted Kaplan-Meier survival estimator (Lumley, 2010) with weight $P(u_i = 1)$. In practice, $\hat{r}_i^u = -\log \hat{S}_u(t_i|x_i)$ and the weight $P(u_i = 1)$ is estimated by w_i in (2.14). The estimated cumulative hazard function can be compared with the 45° line as the usual way for Cox-Snell residuals to examine the goodness-of-fit of the latency submodel in the mixture cure model.

To assist the comparison between the estimated distribution of modified Cox-Snell residuals (with its estimated cumulative distribution function denoted as $\hat{F}_n^{CS}(t)$) and the unit exponential distribution (with the cumulative distribution function denoted as $F_0^{CS}(t) = 1 - e^{-t}$), the Cramér–von Mises criterion (Anderson, 1962) may be used to quantify the distance between the two distributions: $D(\hat{F}_n^{CS}) = \int_0^\infty \left[\hat{F}_n^{CS}(t) - F_0^{CS}(t)\right]^2 dF_0^{CS}(t)$. A smaller $D(\hat{F}_n^{CS})$ indicates a better approximation of the distribution of the modified Cox-Snell residuals by the unit exponential distribution.

3.5.3 Assessing Cure Rate Prediction

The residual approaches discussed above are either not designed to check the model fit of the incidence submodel or less sensitive to departures in the incidence submodel. To examine the performance of the incidence submodel,

one can directly compare the cure status y with the predicted value of $\pi(\boldsymbol{z})$ from a cure model. Let $\hat{\pi}_{d_n}(\boldsymbol{z})$ be the estimator from a cure model based on data $d_n = \{(t_i, \delta_i, y_i, \boldsymbol{x}_i, \boldsymbol{z}_i), i = 1, \ldots, n\}$, and $\{(t_j, \delta_i, y_j, \boldsymbol{x}_j, \boldsymbol{z}_j), j = 1, \ldots, m\}$ are data from m new subjects. The so-called Brier score (Brier, 1950) to measure the discrepancy between y_j and $\hat{\pi}_{d_n}(\boldsymbol{z}_j)$, $j = 1, \ldots, m$ is given by

$$\tilde{e}(\hat{\pi}_{d_n}) = \frac{1}{m} \sum_{j=1}^{m} \{y_j - \hat{\pi}_{d_n}(\boldsymbol{z}_j)\}^2. \tag{3.48}$$

In practice, the cure status is only known for a subject either with $\delta = 1$ and thus uncured ($y = 1$), or with $\delta = 0$ but $t \geq \tau$ and thus cured ($y = 0$). For a subject with $\delta = 0$ and $t < \tau$, the cure status is unknown and cannot be used in (3.48). To correct biases due to the missing values of some y_j's, an inverse probability of censoring weights (IPCW) estimator of $\tilde{e}(\hat{\pi}_{d_n})$ can be calculated (Jiang et al., 2017; Gerds and Schumacher, 2006):

$$\tilde{e}(\hat{\pi}_{d_n}, \hat{G}_n) = \frac{1}{m} \sum_{j=1}^{m} \frac{1 - I(t_j < \tau, \delta_j = 0)}{\hat{G}_n(t_j \wedge \tau)} \{y_j - \hat{\pi}_{d_n}(\boldsymbol{z}_j)\}^2, \tag{3.49}$$

where $\hat{G}_n(\cdot)$ is the Kaplan-Meier estimate of the survival function for censoring times based on d_n and $t_j \wedge \tau = \min(t_j, \tau)$. There are three different weights assigned according to the IPCW scheme in (3.49):

- For a subject with $\delta_j = 1$, the assigned weight is $1/\{m\hat{G}_n(t_j)\}$ based on the fact that $C \geq t_j$ with probability $P(C \geq t_j) = G(t_j)$.
- For a subject with $\delta_j = 0$ and $t_j \geq \tau$, the assigned weight is $1/\{m\hat{G}_n(\tau)\}$ based on the fact that $C \geq \tau$ with probability $P(C \geq \tau) = G(\tau)$.
- For a subject with $\delta_j = 0$ and $t_j < \tau$, the cure status is unknown and the assigned weight is 0. This subject is excluded in the calculation of (3.49).

Let $e(\pi) = \mathrm{E}_F\{Y - \pi(\boldsymbol{z})\}^2$, and suppose that $\hat{\pi}_{d_n}(\boldsymbol{z})$ is a consistent estimator of $\pi(\boldsymbol{z})$. It can be shown that $\tilde{e}(\hat{\pi}_{d_n}, \hat{G}_n)$ of (3.49) is a consistent estimator of $e(\pi)$, and asymptotically equivalent to $\tilde{e}(\hat{\pi}_{d_n})$ of (3.48). The property holds when the censoring distribution depends on covariates $\tilde{\boldsymbol{x}}$. In this case, the survival estimate $\hat{G}_n(\cdot)$ needs to be replaced by the conditional estimate $\hat{G}_n(\cdot|\tilde{\boldsymbol{x}}_j)$, which can be obtained using the method discussed in Section 3.3.1.2, a kernel-type nonparametric regression (Dabrowska, 1987, 1989) for discrete and continuous covariates, or by a grouped Kaplan-Meier estimate for discrete covariates.

Calculating $\tilde{e}(\hat{\pi}_{d_n}, \hat{G}_n)$ in (3.49) requires a large independent test set from new subjects. In practice, however, the large independent test set is usually not available. In this case, one can use the resubstitution method to estimate $\tilde{e}(\hat{\pi}_{d_n}, \hat{G}_n)$ by using d_n both as the training set to build the cure model and

as the test set to estimate $\tilde{e}(\hat{\pi}_{d_n}, \hat{G}_n)$:

$$\hat{e}^{\mathrm{rs}}(\hat{\pi}_{d_n}, \hat{G}_n) = \frac{1}{n}\sum_{i=1}^{n} \frac{1 - I(t_i < \tau, \delta_i = 0)}{\hat{G}_n(t_i \wedge \tau)} \{y_i - \hat{\pi}_{d_n}(\boldsymbol{z}_i)\}^2.$$

The resubstitution method tends to have $\hat{e}^{\mathrm{rs}}(\hat{\pi}_{d_n}, \hat{G}_n)$ that is smaller than $\tilde{e}(\hat{\pi}_{d_n}, \hat{G}_n)$. A better approach is the cross-validation method that splits the observed data set d_n into K folds, denoted by $f_1, \ldots, f_K$, with approximately the same number of subjects, and leaves one fold out as the test set at a time while taking the remaining $K-1$ folds as the training set. The K-fold cross-validation estimate takes the following form

$$\hat{e}^{\mathrm{Kcv}}(\hat{\pi}_{d_n}, \hat{G}_n) = \frac{1}{n}\sum_{k=1}^{K}\sum_{l \in f_k} \frac{1 - I(t_l < \tau, \delta_l = 0)}{\hat{G}_n(t_l \wedge \tau)} \{y_l - \hat{\pi}_{f_{-k}}(\boldsymbol{z}_l)\}^2,$$

where $\hat{\pi}_{f_{-k}}(\boldsymbol{z}_l)$ is an estimator of π based on the observed data set d_n with f_k removed. In this method, $\hat{G}_n(\cdot)$ can be estimated based on all training data d_n instead of all but f_k fold in estimating the IPCW weights because the former is expected to provide more accurate estimates of the weights than the latter.

3.5.4 Concordance Measures for Cure Models

Concordance measures are often used to assess a model fit. For example, area under the curve (AUC) is a concordance measure for logistic regression. Consider the logistic regression in the incidence submodel of the mixture cure model. If all y_i's are available, the AUC is defined as the area under the receiver operating characteristic (ROC) curve, which is a plot of sensitivity

$$\frac{\sum_{i=1}^{n} I(\hat{\pi}(\boldsymbol{z}_i) \geq c) y_i}{\sum_{i=1}^{n} y_i}$$

against 1− specificity

$$\frac{\sum_{i=1}^{n} I(\hat{\pi}(\boldsymbol{z}_i) \geq c)(1 - y_i)}{\sum_{i=1}^{n} (1 - y_i)}$$

for all possible values of c. However, this measure cannot be used directly for the logistic regression in the incidence submodel of the mixture cure model due to the unknown cure status y_i for some subjects. A workaround approach is to replace y_i's by $\hat{\pi}(\boldsymbol{z}_i)$'s for all subjects or only for those subjects with unknown y_i's (Asano et al., 2014). The bootstrap method can be used to estimate the standard error of the AUC, which allows inferences based on the AUC.

A concordance measure can also be defined for the failure times. Zhang and Shao (2017) considered a slightly more general transformation model than (3.12) for the latency part of the mixture cure model

$$S_u(t|\boldsymbol{x}) = S_\epsilon[g(t) + \phi(\boldsymbol{x}'\boldsymbol{\beta})],$$

where $\phi(\cdot)$ is a known function, and suggested a k-index concordance measure

$$K(\hat{\boldsymbol{\beta}}, \hat{\boldsymbol{\gamma}}) = \frac{\sum_{i \neq j} G(\boldsymbol{x}_i'\hat{\boldsymbol{\beta}}, \boldsymbol{x}_j'\hat{\boldsymbol{\beta}}) I(\boldsymbol{x}_i'\hat{\boldsymbol{\beta}} < \boldsymbol{x}_j'\hat{\boldsymbol{\beta}}) \hat{\pi}(\boldsymbol{z}_i)\hat{\pi}(\boldsymbol{z}_j)}{\sum_{i \neq j} I(\boldsymbol{x}_i'\hat{\boldsymbol{\beta}} < \boldsymbol{x}_j'\hat{\boldsymbol{\beta}}) \hat{\pi}(\boldsymbol{z}_i)\hat{\pi}(\boldsymbol{z}_j)}$$

where

$$G(u_1, u_2) = \int_{-\infty}^{u_2} [S_\epsilon(u_1 - u_2 + u) - S_\epsilon(u_1)] dS_\epsilon(u)$$

This measure extends Harrell's c-index for PH model and can be shown to be asymptotically normal. A bootstrap method can be employed to estimate the standard error of the estimate of the measure, which allows inferences based on the k-index. A similar measure can also be defined for the transformation model for the non-mixture cure model discussed in Section 2.4.4.

3.5.5 Testing Goodness-of-Fit of Parametric Cure Rate Estimation

The proposed nonparametric estimation method for $\pi(z)$ in Section 3.3.1 is flexible to accommodate different effects of z on π than the parametric estimation methods based on logistic models or parametric models based on the complementary log-log link function discussed in Section 2.2.1. However, the parametric methods are more efficient if the parametric assumptions are correct and the results from the parametric methods are easier to interpret. Therefore, it is of interest to examine the goodness-of-fit of the parametric estimates for $\pi(z)$ relative to the flexible nonparametric estimate for $\pi(z)$ to determine any significant difference between them. This is similar to the question about comparing nonparametric versus parametric regression fits (Hardle and Mammen, 1993).

Let $\hat{\pi}(z)$ be the nonparametric estimate in (3.33) and $\pi_{\hat{\boldsymbol{\gamma}}}(z)$ be the parametric estimate in (2.2) or (2.3). The hypotheses for the goodness-of-fit question can be expressed as follows:

$$H_0\colon \pi(z) = \pi_{\boldsymbol{\gamma}}(z) \text{ for some } \boldsymbol{\gamma} \in \boldsymbol{\Gamma} \text{ vs } H_1\colon \pi(z) \neq \pi_{\boldsymbol{\gamma}}(z) \text{ for all } \boldsymbol{\gamma} \in \boldsymbol{\Gamma} \quad (3.50)$$

where $\boldsymbol{\Gamma}$ is a finite-dimensional parameter space of $\boldsymbol{\gamma}$ and the function $\pi_{\boldsymbol{\gamma}}(z)$ is a known function up to a parameter vector $\boldsymbol{\gamma} \in \boldsymbol{\Gamma}$. A goodness-of-fit test to test above hypotheses can be constructed based on the following distance measure between the two estimates (Müller and van Keilegom, 2019)

$$\mathcal{T}_n = nh^{1/2} \int \{\hat{\pi}(z) - \pi_{\hat{\boldsymbol{\gamma}}}(z)\}^2 p(z) dz$$

or its empirical version

$$\tilde{\mathcal{T}}_n = nh^{1/2} \frac{1}{n} \sum_{i=1}^{n} \{\hat{\pi}(z_i) - \pi_{\hat{\boldsymbol{\gamma}}}(z_i)\}^2,$$

where $p(z)$ is the density function of z and h is the bandwidth used in the nonparametric estimate $\hat{\pi}(z)$. Let $H(t|z) = P(\min(T,C) \leq t|z)$, $H_1(t|z) = P(\min(T,C) \leq t, \delta = 1|z)$, $\zeta(t|z) = \frac{I(T \leq t, T \leq C)}{1-H(\min(T,C)|z)} - \int_0^t \frac{I(\min(T,C) \geq s)}{[1-H(s|z)]^2} dH_1(s|z)$, $\tau_u(z) = \inf\{t\colon S_u(t|z) = 0\}$ and $\tau_G(z) = \inf\{t\colon G(t|z) = 0\}$. Under the sufficient follow-up condition $\tau_u(z) < \tau_G(z)$ and some mild conditions for $p(z)$, $\pi(z)$, the kernel function in the nonparametric estimate $\hat{\pi}(z)$, the latency survival time distribution and the censoring time distribution, if $nh^3(\log n)^{-5} \to \infty$ and $nh^5/(\log n) = O(1)$ as $n \to \infty$, the two distance measures are asymptotically equivalent and $\mathcal{T}_n$ is asymptotically normally distributed under the null hypothesis in (3.50) with

$$\mathcal{T}_n - b_h \to N(0, V),$$

where $b_h = h^{-1/2} \int K^2(s)ds \int_0^1 [1 - \pi(z)]^2 \mu(z)dz = O(h^{-1/2})$, $V = 2K^{(4)}(0) \int_0^1 [(1-\pi(z))^2 \mu(z)]^2 dz$, $K^{(4)}$ denotes the fourth convolution product of K and $\mu(z) = E[\zeta^2(\tau_u(z)|z)|Z = z]$. Under a local alternative of the form $\pi(z) = \pi_{\boldsymbol{\gamma}}(z) + n^{-1/2}h^{-1/4}\Delta_n(z)$ with $\Delta_n(z)$ bounded uniformly in z and n, $\mathcal{T}_n$ is also asymptotically normally distributed

$$\mathcal{T}_n - b_h - \int \Delta_n(z)^2 p(z)dz \to N(0, V),$$

which enables power and sample size estimation. However, due to slow convergence rates, the normal distribution does not provide an accurate approximation to the null distribution of $\mathcal{T}_n$ or $\tilde{\mathcal{T}}_n$ under finite sample sizes, and a bootstrap method is suggested to better approximate the distribution (Müller and van Keilegom, 2019). A proposed algorithm for the bootstrap method to approximate the null distribution is given below:

Step 1 Use the data, a pilot bandwidth h_0 and the estimator in Section 3.3.1 to obtain $\hat{\pi}_{h_0}(z_i)$, $\hat{S}_{h_0}(t|z_i)$ and $\hat{G}_{h_0}(t|z_i)$.

Step 2 Fit a logistic model to data $\{(z_i, \hat{\pi}_{h_0}(z_i)), i = 1, \ldots, n\}$ to obtain $\hat{\boldsymbol{\gamma}}$. That is, $\hat{\boldsymbol{\gamma}} = \arg\max_{\boldsymbol{\gamma}} \sum_{i=1}^n [\hat{\pi}_{h_0}(z_i) \log(\pi_{\boldsymbol{\gamma}}(z_i) + (1 - \hat{\pi}_{h_0}(z_i)) \log(1 - \pi_{\boldsymbol{\gamma}}(z_i))]$. The corresponding parametric estimator is denoted as $\hat{\pi}_{h_0,\hat{\boldsymbol{\gamma}}}(z_i)$.

Step 3 Generate the bth of B bootstrap samples data $\{(z_i, t^*_{i,b}, \delta^*_{i,b}), i = 1 \ldots, n\}$ from the mixture cure model with the latency uncured survival function $[\hat{S}_{h_0}(t|z_i) - 1 + \hat{\pi}_{h_0}(z_i)]/\hat{\pi}_{h_0}(z_i)$ and incidence uncure rate $\hat{\pi}_{h_0,\hat{\boldsymbol{\gamma}}}(z_i)$. The censoring distribution is set to $\hat{G}_{h_0}(t|z_i)$.

Step 4 For the bth bootstrap sample and a bandwidth h, use the method of Section 3.3.1 to obtain a nonparametric estimate $\hat{\pi}^*_{hh_0,b}(z_i)$, and then use $(z_i, \hat{\pi}^*_{hh_0,b}(z_i))$ and the method in Step 2 to obtain a parametric estimate $\hat{\pi}_{hh_0,\hat{\boldsymbol{\gamma}}^*,b}(z_i)$.

Step 5 Find the $(1-\alpha)\times 100$th percentile of $\{\mathcal{T}^*_{n,b}, b = 1, \ldots, B\}$ as the critical value of the test, where

$$\mathcal{T}^*_{n,b} = nh^{-1/2}\frac{1}{n}\sum_{i=1}^{n}\left[\hat{\pi}^*_{hh_0,b}(z_i) - \hat{\pi}_{hh_0,\hat{\boldsymbol{\gamma}}^*,b}(z_i)\right]^2.$$

3.5.6 Variable Selection

Similar to other regression models, variable selection is an important issue in the mixture cure model, particularly when there are a large number of covariates available for the two regression submodels involved in a mixture cure model. The AIC discussed in Section 2.5.1 has long been used to select a parsimonious model among candidate models. For a parametric mixture cure model with observed maximum likelihood function (2.12) available, it is not difficult to use this method to select covariates and models in the mixture cure model context, as discussed in Section 2.5.1.

For semiparametric mixture cure models, since most of the models are estimated using the EM algorithm, the observed maximum likelihood function is not readily available, which makes AIC calculation difficult. Dirick et al. (2015) proposed an approach to calculate the AIC values based on the maximum complete likelihood function (2.15) and (2.16).

A popular and modern approach to select covariates in regression models is via penalization, such as LASSO (Tibshirani, 1996) or SCAD (Fan and Li, 2001) methods. We demonstrate in this section how these methods can be implemented for mixture cure models under the EM framework (Liu et al., 2012). Consider the semiparametric PH model in Section 3.2.1. If the EM algorithm is used to estimate the parameters in the model, then the M-step maximizes (2.15) and (3.2). The penalized method suggests replacing the two functions by their corresponding penalized versions:

$$\ell_1^P(\boldsymbol{\gamma}) = \log\prod_{i=1}^{n}\left\{\pi(\boldsymbol{z}_i)^{w_i^{(k)}}[1-\pi(\boldsymbol{z}_i)]^{1-w_i^{(k)}}\right\} - n\sum_{j=1}^{p_z}P_{\lambda_{1j}}(|\gamma_j|) \tag{3.51}$$

$$\ell_2^P(\boldsymbol{\beta},\boldsymbol{\alpha}) = \log\prod_{i=1,w_i^{(k)}>0}^{n}\left\{h_u(t_i|\boldsymbol{x}_i)^{\delta_i}S_u(t_i|\boldsymbol{x}_i)^{w_i^{(k)}}\right\} - n\sum_{j=1}^{p_x}P_{\lambda_{2j}}(|\beta_j|) \tag{3.52}$$

where $P_\lambda(|\cdot|)$ is a penalty function and $\boldsymbol{\lambda} = (\lambda_{11}, \ldots, \lambda_{1p_z}, \lambda_{21}, \ldots, \lambda_{1p_x})$ are tuning parameters, which may be chosen by a data-driven method, such as generalized cross validation (GCV). The penalty function can be either the LASSO penalty $P_\lambda(|x|) = \lambda|x|$ or the SCAD penalty

$$P_\lambda(|x|) = \begin{cases} \lambda|x| & \text{if } 0 \le |x| < \lambda \\ \frac{(a^2-1)\lambda^2-(|x|-a\lambda)^2}{2(a-1)} & \text{if } \lambda \le |x| < a\lambda \\ \frac{(a+1)\lambda^2}{2} & \text{if } |x| \ge a\lambda. \end{cases}$$

The penalized likelihood (3.51) can be maximized by the Newton-Raphson method or by a program that can handle penalized logistic regression analysis. The penalized likelihood (3.52) can be replaced with a penalized partial likelihood discussed in (Fan and Li, 2002) to obtain an update of $\boldsymbol{\beta}$ and the baseline survival function can be estimated by the Nelson-Aalen estimator (3.3). Since the added penalty terms do not depend on y_i, the E-step does not change and (2.14) is calculated to complete the E-step.

To speed up the estimation, instead of minimizing GCV with respect to the $p_x + p_z$-dimensional parameter $\boldsymbol{\lambda}$, one can set $\lambda_{1j} = \lambda_1 \mathrm{SE}(\hat{\gamma}_j^u)$ and $\lambda_{2j} = \lambda_2 \mathrm{SE}(\hat{\beta}_j^u)$ where λ_1 and λ_2 are tuning parameters and $\mathrm{SE}(\hat{\gamma}_j^u)$ and $\mathrm{SE}(\hat{\beta}_j^u)$ are the standard errors of the estimates from semiparametric PH model in Section 3.2.1 without the penalties. The tuning parameters λ_1 and λ_2 can be selected by minimizing GCVs from the logistic regression model and the Cox's PH model in the M-step. If some variable is believed to be or must be in the model, the corresponding tuning parameter in $\boldsymbol{\lambda}$ can be set to zero, that is, the penalty on the variable's coefficient is zero. For example, we do not put any penalty on the intercept because the intercept is always in the logistic regression part.

A similar idea can also be applied to other semiparametric cure models, such as the semiparametric PHC model (Masud et al., 2018). For parametric cure models, due to the availability of the observed likelihood function, it is possible to penalize the observed likelihood function directly instead of the complete likelihood function in the M-step (Scolas et al., 2016).

For semiparametric spline-based additive effects mixture cure model discussed in Section 3.2.6, the penalized likelihood approach above does not work since the effects of covariates are in the form of functionals rather than in a form of coefficients. Variable selection can be carried out using the Kullback-Leibler distance (Gu, 2013). For two estimates ζ_1 and ζ_2 of ζ in (3.24), the Kullback-Leibler distance between them is defined as

$$\mathrm{KL}(\zeta_1, \zeta_2) = \frac{1}{n} \sum_{i=1}^{n} \Bigg[\frac{e^{\zeta_1(\boldsymbol{z}_i)}}{1 + e^{\zeta_1(\boldsymbol{z}_i)}} \{\zeta_1(\boldsymbol{z}_i) - \zeta_2(\boldsymbol{z}_i)\} - \{\log(1 + \zeta_1(\boldsymbol{z}_i)) - \log(1 + \zeta_2(\boldsymbol{z}_i))\} \Bigg]$$

For two estimates η_1 and η_2 of η in (3.25), the Kullback-Leibler distance between them is defined as

$$\mathrm{KL}(\eta_1, \eta_2) = \frac{1}{n} \sum_{i=1}^{n} \int_0^{t_i} \Big[e^{\eta_1(t, \boldsymbol{x}_i)} \{\eta_1(t, \boldsymbol{x}_i) - \eta_2(t, \boldsymbol{x}_i)\} - \{e^{\eta_1(t, \boldsymbol{x}_i)} - e^{\eta_2(t, \boldsymbol{x}_i)}\} \Big].$$

Let $\hat{\zeta}$ be an estimate of ζ in a space $\mathcal{H}_1$ for a full model, $\hat{\zeta}^*$ be an estimate of ζ in a subspace $\mathcal{H}_2 \subset \mathcal{H}_1$ for a reduced model, and $\hat{\zeta}_c$ be an estimate of ζ from

a constant model. Then $\mathrm{KL}(\hat{\zeta}, \hat{\zeta}^*)/\mathrm{KL}(\hat{\zeta}, \hat{\zeta}_c) \leq 0.05$ can be used to favor the reduced model. A similar criterion can be obtained for $\hat{\eta}$ (Wang et al., 2012a).

3.6 Software and Applications

There are several R and SAS packages that can be used to fit the semiparametric and nonparametric cure models. We will show some software packages and use them to illustrate the applications of some cure models discussed in early sections to real data sets.

3.6.1 R Package mixcure

The R package `mixcure` can fit the semiparametric PH mixture cure model in addition to the parametric mixture cure models discussed in Section 2.6.2. We demonstrate the use of this package to fit the model to the leukemia data introduced in Section 2.6 as follows. The treatment group variable `transplant` enters both parts of the mixture cure model.

```
> summary(mixcure(Surv(time, cens) ~ transplant, ~ transplant, data
+ = goldman.data, savedata = T), R = 300)

Call:
mixcure(lformula = Surv(time, cens) ~ transplant, iformula =
    ~transplant, data = goldman.data, savedata = T)

Latency model for uncured:
                     coefficient    stderr   zscore     pvalue
transplantAutologous   0.6332632 0.3085259 2.052545 0.04011672

Incidence model:
                     coefficient    stderr    zscore      pvalue
(Intercept)            1.0283684 0.3703484 2.7767593 0.005490381
transplantAutologous   0.3769905 0.7157321 0.5267202 0.598387918
```

Under this model, the two treatment groups are not statistically significant in cure rates. However, the Autologous group does show significantly higher hazards than the Allogeneic group with an estimated hazard ratio of 1.88 and 95% confidence interval (1.03, 3.45). The fitted survival curves are plotted against the Kaplan-Meier survival curves in Figure 3.2. It is clear that the semiparametric PH mixture cure model provides a better fit to the data than the previous parametric mixture cure models, and the improved fit leads to a significant effect of the treatment in the latency part.

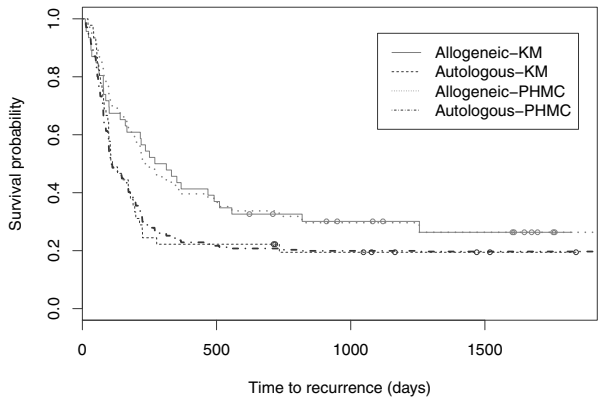

FIGURE 3.2
Estimated survival curves (dotted lines) from the semiparametric PH mixture cure model for the leukemia data.

3.6.2 R Package smcure

The R package `smcure` can also fit the semiparametric PH mixture cure model as follows

```
> smcure(Surv(time, cens) ~ transplant, ~ transplant, data =
+ goldman.data, model = "ph", nboot = 500)

Call:
smcure(formula = Surv(time, cens) ~ transplant, cureform =
    ~transplant, data = goldman.data, model = "ph", nboot = 500)

Cure probability model:
             Estimate Std.Error   Z value  Pr(>|Z|)
(Intercept) 0.6986655 0.8792946 0.7945750 0.4268608
transplant  0.3579095 0.6011734 0.5953516 0.5516085

Failure time distribution model:
                      Estimate Std.Error  Z value   Pr(>|Z|)
transplantAutologous 0.6363645 0.3314896 1.919712 0.05489429
```

There are some differences in the estimates of incidence parameters between `mixcure` and `smcure`. The differences may be due to the quasi-binomial-logit link instead of the binomial-logit link used in `smcure`.

Another model `smcure` can fit is the semiparametric AFT mixture cure model discussed in Section 3.2.2. We can use it to fit the leukemia data as follows:

```
> smcure(Surv(time, cens) ~ transplant, ~ transplant, data =
+ goldman.data, model = "aft", nboot = 500)
```

```
Call:
smcure(formula = Surv(time, cens) ~ transplant, cureform =
    ~transplant, data = goldman.data, model = "aft", nboot = 500)

Cure probability model:
             Estimate Std.Error   Z value  Pr(>|Z|)
(Intercept) 0.5800933 0.7795188 0.7441684 0.4567746
transplant  0.4273070 0.5286190 0.8083459 0.4188915


Failure time distribution model:
                       Estimate Std.Error   Z value  Pr(>|Z|)
(Intercept)           0.2101563 0.1872494  1.122333 0.2617208
transplantAutologous -0.3531250 0.2706421 -1.304768 0.1919720
```

The results show that the two treatment groups are not statistically significant in either of the incidence and latency part. The differences in results between the semiparametric PH model and the semiparametric AFT model may partly be due to the poor fit of the AFT model to the data.

3.6.3 SAS Macro PSPMCM

The SAS macro `PSPMCM` (Corbiere and Joly, 2007) can fit the semiparametric PH model as follows

```
%pspmcm(DATA=goldmandata,ID=id,CENSCOD=cens,TIME=time,
  VAR= transplant(IS), INCPART=logit, SURVPART=cox,
  TAIL=zero, SU0MET=pl, FAST=Y,BOOTSTRAP=Y,
  NSAMPLE=2000, STRATA=, MAXITER=200,CONVCRIT=1e-5,
  ALPHA=0.05, BASELINE=Y, BOOTMET=ALL, JACKDATA=,
  GESTIMATE=Y, SPLOT=Y, PLOTFIT=Y);
run;
```

```
                        FAST RESULTS FOR THE LOGIT PART (goldmandata)
                           Analysis of Maximum Likelihood Estimates

                                 Standard     Wald        Pr >     Estimation
       Variable   DF  Estimate   Error    Chi-Square  Chi-Square     Type
      Intercept    1    1.0283    0.3347      9.4373    0.0021       MLE
      transplant   1    0.3770    0.5025      0.5629    0.4531       MLE

                       FAST RESULTS FOR THE SURVIVAL PART (goldmandata)
                           Analysis of Maximum Likelihood Estimates

                                                                  95% Lower
                                                                 Confidence
             Parameter  Standard               Pr >  Hazard    Limit for
Parameter  DF  Estimate     Error Chi-Square   ChiSq   Ratio Hazard Ratio
transplant  1   0.63323   0.26031     5.9175  0.0150   1.884     1.131
lpi         0   1.00000         0      .       .        .         .

             95% Upper
            Confidence
             Limit for
Parameter  Hazard Ratio
```

```
transplant     3.138
lpi             .

                    BOOTSTRAP CONFIDENCE INTERVAL FOR PARAMETERS ESTIMATES
                      (confidence level=95%, 2000 bootstrap resamples)

                    Method=BC  Method=BC Method=BCA Method=BCA
                      Lower      Upper      Lower      Upper     Method=    Method=
          Observed Confidence Confidence Confidence Confidence PCTL Lower PCTL Upper
Variable Statistic    Limit      Limit      Limit      Limit   percentile percentile
L_Int     1.02808    0.35720    1.89458    0.33647    1.83466    0.34868    1.85686
L_TRANSP  0.37722   -0.82064    1.46265   -0.86845    1.44908   -0.67219    1.56175
S_TRANSP  0.63323    0.03407    1.26047   -0.02173    1.22178    0.05280    1.30678

          Method=HYB
             Lower
Variable  percentile
L_Int       0.19930
L_TRANSP   -0.80732
S_TRANSP   -0.04032

                                Method=BOOTN Method=BOOTN Method=JACK Method=JACK
                   Method=HYB      Lower        Upper       Lower       Upper
            Observed    Upper     Confidence   Confidence   Confidence  Confidence
Variable   Statistic percentile      Limit        Limit       Limit       Limit
L_Int       1.02808    1.70748      0.26070      1.76367      0.27630    1.72789
L_TRANSP    0.37722    1.42663     -0.77328      1.44392     -0.71742    1.40118
S_TRANSP    0.63323    1.21366     -0.00909      1.23399     -0.05167    1.27977
```

The results are very similar to the results from `mixcure`.

The SAS macro `PSPMCM` also provides `WTAIL` and `ETAIL` options in addition to `ZERO` when estimating the upper tail of the baseline distribution in the latency submodel. See more detailed discussions of the tail completion methods in Section 3.2.1.1.

3.6.4 R Package Survival

The R package `survival` can be used to fit a mixture cure model and estimate cure rate nonparametrically with one categorical variable or discrete variable using the method discussed in Section 3.3.1. Using the leukemia data introduced in Section 2.6 as an example, we first obtain the nonparametric survival estimates of the two treatment groups using `survfit` function in `survival` package as follows:

```
> z = survfit(Surv(time, cens) ~ transplant, data = goldman.data)
> summary(z)

Call: survfit(formula = Surv(time, cens) ~ transplant,
      data = goldman.data)

                 transplant=Allogeneic
 time n.risk n.event survival std.err lower 95% CI upper 95% CI
   11     46       1    0.978  0.0215        0.937        1.000
   14     45       1    0.957  0.0301        0.899        1.000
   23     44       1    0.935  0.0364        0.866        1.000
    .      .       .        .       .            .            .
```

```
    .      .       .        .       .            .            .
    .      .       .        .       .            .            .
  557     16       1    0.326  0.0691        0.215        0.494
  819     13       1    0.301  0.0682        0.193        0.469
 1256      8       1    0.263  0.0693        0.157        0.441

                transplant=Autologous
 time n.risk n.event survival std.err lower 95% CI upper 95% CI
   21     45       1    0.978  0.0220        0.936        1.000
   40     44       1    0.956  0.0307        0.897        1.000
   42     43       1    0.933  0.0372        0.863        1.000
    .      .       .        .       .            .            .
    .      .       .        .       .            .            .
    .      .       .        .       .            .            .
  224     12       1    0.244  0.0641        0.146        0.409
  277     11       1    0.222  0.0620        0.129        0.384
  734      8       1    0.194  0.0601        0.106        0.356
```

Following (3.30), the cure rate estimates for the two treatment groups are given by the last lines of the survival output for the two groups as follows

```
> summary(z[1], time = 1256)

Call: survfit(formula = Surv(time, cens) ~ transplant,
      data = goldman.data)

 time n.risk n.event survival std.err lower 95% CI upper 95% CI
 1256      8      33    0.263  0.0693        0.157        0.441
> summary(z[2], time = 734)
Call: survfit(formula = Surv(time, cens) ~ transplant,
      data = goldman.data)

 time n.risk n.event survival std.err lower 95% CI upper 95% CI
  734      8      36    0.194  0.0601        0.106        0.356
```

That is, the cure rate and its 95% confidence intervals are 0.263 and (0.157, 0.441) for Allogeneic group and 0.194 and (0.106, 0.356) for Autologous group. The latency survival functions of the two groups can be obtained using the method of (3.37). For example, the estimated latency survival function for the Allogeneic group is

```
> cbind(time = z[1]$time, surv = (z[1]$surv - cure1)/(1-cure1))

      time       surv
 [1,]   11 0.97048808
 [2,]   14 0.94097616
 [3,]   23 0.91146425
    .    .          .
    .    .          .
    .    .          .
```

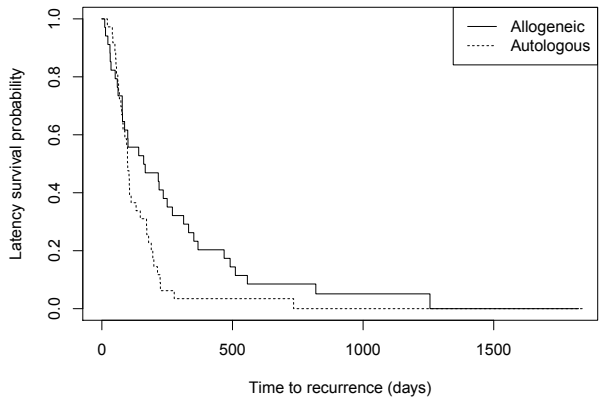

FIGURE 3.3
Estimated latency survival curves from the nonparametric mixture cure model for the leukemia data.

```
[36,] 1109 0.05107832
[37,] 1158 0.05107832
[38,] 1256 0.00000000
```

Similar estimation can be done for Autologous group. The estimated survival curves can be obtained as follows and plotted in Figure 3.3:

```
> cure1 = summary(z[1], time = 1256)$surv
> cure2 = summary(z[2], time = 734)$surv
> plot(cbind(time = c(0, z[1]$time), surv = c(1, (z[1]$surv -
+ cure1)/(1-cure1))), type = "s", ylab = "Latency survival
+ probability", xlab = "Time to recurrence (days)")
> lines(cbind(time = c(0, z[2]$time), surv = c(1, (z[2]$surv -
+ cure2)/(1-cure2))), type = "s", lty = 2)
> legend("topright", legend = c("Allogeneic","Autologous"), lty=1:2)
```

3.6.5 R Package npcure

To fit a nonparametric cure rate model with a continuous covariate, one can use an R package `npcure`. We consider the data from a study of bone marrow transplant for leukemia patients reported by Copelan et al. (1991) (the data is available in Klein and Moeschberger (2003)). The study included 137 leukemia patients from hospitals in the United States and Australia, and they were followed up to 2,640 days. There are 42 patients relapsed, 41 died in remission and 54 censored at the end of the study (the overall censoring percentage was 39.4%). The event of interest is relapse or death due to leukemia following bone marrow transplantation. Figure 3.4 shows the Kaplan-Meier survival function estimate for the data. The curve shows a long plateau that levels off

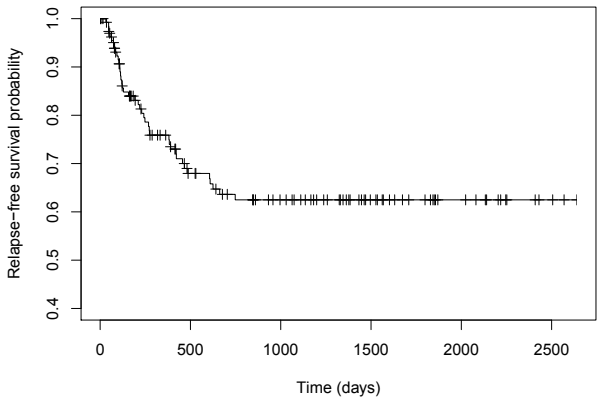

FIGURE 3.4
The Kaplan-Meier survival curve of time to relapse or death due to leukemia after bone marrow transplant.

substantially above zero. It indicates that a substantial fraction of patients may be viewed as cured after a sufficiently long follow-up, and cure models may be suitable to analyze the data.

Following the work of Xu and Peng (2014), we consider patient's age effect in the cure model in the following analysis. The data are stored in `bmt`.

```
> str(bmt)

'data.frame':   137 obs. of  4 variables:
 $ time: int  2081 1602 1496 1462 1433 1377 1330 996 226 1199 ...
 $ cens: int  0 0 0 0 0 0 0 0 0 0 ...
 $ pAge: int  26 21 26 17 32 22 20 22 18 24 ...
 $ dAge: int  33 37 35 21 36 31 17 24 21 40 ...
```

The function `probcure` in `npcure` can be used to obtain a nonparametric estimate of the effect of `pAge` on cure rate and its 95% confidence interval as follows. A plot of the estimated effect and its confidence interval is given in Figure 3.5. The figure is slightly different from Figure 3 in Xu and Peng (2014) due to local optimal bandwidths are used at different values of `pAge`.

```
> z = probcure(pAge, time, cens, data = bmt, x0 = 7:52,   conflevel
+ = 0.95)
> plot(with(z, cbind(x0, q)), type = "l", ylim = c(0, 1), xlab =
+ "Patient's age", ylab = "Cure rate")
> lines(with(z, cbind(x0, conf$lower)), lty = 2)
> lines(with(z, cbind(x0, conf$upper)), lty = 2)
```

The package also provides a function `latency` to estimate the latency survival function using the method of Section 3.3.2. We use the function to estimate the latency survival functions for patient's age from 7 to 52 and plot them in Figure 3.6 as follows.

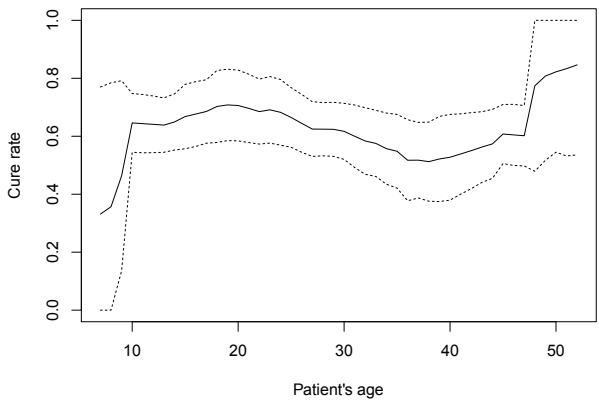

FIGURE 3.5
The estimated patient's age effect on cure rate and its 95% confidence interval for leukemia patients after bone marrow transplant.

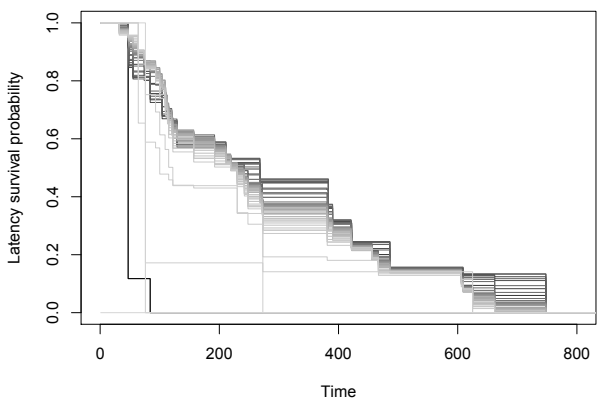

FIGURE 3.6
The estimated latency survival functions for leukemia patients with age 7 (darkest line) to 52 (lightest line) after bone marrow transplant.

```
> z = latency(pAge, time, cens, data = bmt, x0 = 7:52)
> plot(with(z, cbind(testim, S[[1]])), type = "s", ylim = c(0, 1),
+ xlab = "Time", ylab = "Latency survival probability", xlim = c(0,
+ 800), col = gray.colors(1))
> lapply(1:length(z$S), function(x)lines(z$testim, z$S[[x]], type =
+ "s", col = gray.colors(length(z$S), start = 0, end = 0.9)[x]))
```

The figure shows that patient's age has some effects on the survival within 100 days and between 200 and 400 days. However, the survival curves cross and show some potential complicated effect of patient's age on the survival time of the uncured patients, which indicates a parametric or semiparametric latency submodel, such as the PH model, may not be appropriate for modeling such an effect.

3.7 Summary

Semiparametric cure models are the models with unspecified baseline distributions that are usually estimated by a nonparametric method. Other parts of the models are parametrically specified, such as the incidence submodel and the covariate effects in the latency submodel of the mixture cure models. This chapter presented details of the semiparametric mixture cure models under various assumptions in the latency submodels and their estimation methods. In particular, we demonstrated that the EM algorithm can simplify the estimation process and connect the cure models with standard cure models. We also showed the semiparametric non-mixture cure models and their estimation methods. Cure models without any parametric assumptions are referred to as nonparametric cure models. We introduced two methods to estimate the incidence submodel and the latency submodel under the nonparametric mixture cure model based on the generalized Kaplan-Meier estimator. We also discussed some model diagnosis methods based on mixture cure models, including residuals, good-of-fit measures, and variable selection using the penalized complete likelihood functions in the EM algorithm. Finally, we demonstrated 4 R packages and 1 SAS macro to fit some semiparametric cure models.

We skipped some details of the asymptotic properties of the estimates of the semiparametric and nonparametric cure models. For semiparametric mixture cure models, we refer readers to Fang et al. (2005), Lu and Ying (2004), and Zeng et al. (2006) among others for further details.

One consequence of using semiparametric and nonparametric cure models versus parametric cure models is that the nonparametric latency distribution estimate cannot be extrapolated beyond the last observed data, which results in an identifiability issue. We introduced a few methods in this chapter to overcome the identifiability issue. One popular remedy is to set the estimated latency survival function to zero after the largest uncensored time, which

essentially assumes that the largest uncensored time is the threshold for cure. The single cure threshold for every subject may not applicable in other situations, and Betensky and Schoenfeld (2001) proposed a method to deal with random cure threshold. Wu et al. (2014) further considered a situation where the cure status can be detected with a diagnostic procedure with prespecified sensitivity and specificity and adapted the EM algorithm for the semiparametric mixture cure model to include the information in the estimation.

We only considered covariate effects in the latency submodel in the semi parametric cure models via hazard functions or the mean of log survival time in this chapter. Semiparametric mixture cure models with quantile regression as the latency submodel have also been proposed, where the covariate effects in the latency submodel is via the quantile of the latency distribution. See Wu and Yin (2013) and Wu and Yin (2017a,b) for details.

Covariates with measurement errors or mismeasured covariates and partially missed covariates in cure models are also well studied in the literature. In the context of semiparametric and nonparametric cure models, we refer readers to Bertrand et al. (2017) and Beesley et al. (2016) for details.

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